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Research Paper

A general approach for mixed-integer predictive control of HVAC systems using MILP

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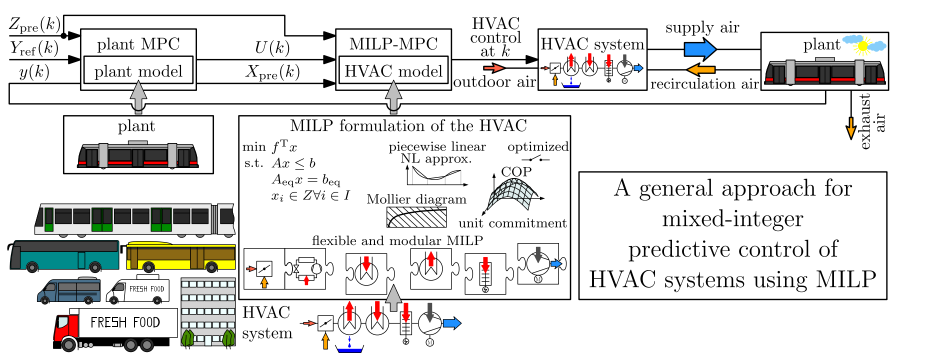
HIGHLIGHTS

- Optimized HVAC operation highly versatile in terms of energy, comfort, and wear.
- Modular and flexible MILP model of HVACs to achieve (near) global optimum.
- Hierarchical MIMO model predictive control (MPC) framework for HVAC operation.
- Adjustable piecewise-linear approximation of nonlinear component characteristics.
- Optimal switching decision for components with dwell times.

GRAPHICAL ABSTRACT



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ABSTRACT

The topic of the paper is optimal control of heating, ventilation, and air conditioning (HVAC) systems. Using mixed-integer linear programming (MILP) the main contribution is a flexible and modular MILP model of HVACs. It is the centerpiece of the proposed hierarchical multi-input multi-output (MIMO) model predictive control (MPC) framework for energy, comfort, and wear optimization. On the upper-level a plant MPC takes care of the slow dynamics of a plant. On the lower-level a MILP-MPC based on a MILP optimization model of an HVAC system optimizes the HVAC operation. The MILP-MPC considers the first few samples of the control trajectory of the plant MPC as its reference trajectory. It minimizes power consumption and switching of the HVAC. Thereby it has to obey constraints, consider component characteristics by nonlinearity approximation, and solve a unit commitment problem. The fact that the plant MPC can be based on an almost arbitrary plant model and the MILP model of the MILP-MPC can represent a variety of different HVAC systems unifies HVAC control. Features and results are presented in case studies: The first case study shows that state of the art HVAC operation is outperformed even if the predictive capability is not used. The second case study demonstrates that temperature, CO_2 level, and humidity can be controlled simultaneously in a decoupled fashion. The third case study reveals that if also the lower-level MILP-MPC oversees the latency periods of HVAC components optimal switching is achieved.

Abbreviations: AC, air condition; CHP, combined heat and power; COP, coefficient of performance; DOF, degrees of freedom; FSM, finite state machine; HB, heater battery; HEN, heat exchanger network; HP, heat pump; HVAC, heating, ventilation, and air conditioning; IDA, indoor air; LP, linear programming; MILP, mixed-integer linear programming; MIMO, multi-input multi-output; MINLP, mixed-integer nonlinear programming; MPC, model predictive control; NLP, nonlinear programming; ODA, outdoor air; PID, proportional integral derivative; PWM, pulse width modulation; QP, quadratic program; RBF, radial basis function; RCA, recirculation air; RM, refrigeration machine; SAF, supply air fan; SOS-2, special ordered set of type 2; SUP, supply air; TES, thermal energy storage; ULF, Ultra Low Floor; VCP, vapor compression plant.

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Nomenclature

$b_{AC\bullet}$	by-pass valve state refrigeration machine $\bullet$
c_A	volume fraction CO_2 air mass/mass flow A
$c_{p,a}/c_{p,v}$	specific heat capacity dry air/water vapor
$\bullet_{max}/\bullet_{min}$	maximal/minimal value of quantity $\bullet$
$\bullet^*$	optimal value of quantity $\bullet$
$\underline{\bullet}$	parameter provided to the MILP
h_A	specific enthalpy wet air mass/mass flow A
$\dot{H}_{H_2O,COND}$	enthalpy flow of condensed water
$h_{H_2O,COND}$	specific enthalpy of condensed water
i	time instance of a prediction
$I_{\bullet,\uparrow^{ON}_{OFF}}$	switching component $\bullet$ on indicator
$J_\bullet$	MILP cost function term of $\bullet$
$J(k)$	overall MILP cost function at time step k
$J_P(k)$	MILP cost function term electrical power
$J_{p,\bullet,change}$	MILP cost function term penalizing state changes of component $\bullet$
$J_{p,\bullet,\uparrow^{ON}_{OFF}}$	MILP cost function term penalizing switching component $\bullet$ on
$J_{p,u_1,\vartheta_{SUP\bullet}}$	MILP cost function term supply air temperature ϑ_{SUP} violations
$J_{p,u_j}(k)$	MILP cost function term penalizing deviations from desired plant input j
k	current time step
$\dot{m}_A$	wet air mass flow A
$\dot{m}_{H_2O,COND}$	water mass flow of condensed water
$\underline{m}_\bullet$	lower bound utilized in big-M formulation
$\underline{M}_\bullet$	upper bound utilized in big-M formulation
n	largest index number (i.e. $j = 1,\dots,n$)
N_c	control horizon plant MPC
$N_{c,MILP}$	control horizon MILP-MPC
N_p	prediction horizon plant MPC
$N_{p,MILP}$	prediction horizon MILP-MPC
$P(k+i)$	power consumption HVAC at $k+i$
$P_\bullet$	electrical power consumption of component $\bullet$
$\boldsymbol{Q}(k+i)$	plant MPC weighting matrix reference tracking
$\dot{Q}_{AC\bullet xb}$	reduced cooling power refrigeration machine (first type; version $\bullet$) due to by-pass
$\dot{Q}_{HP\bullet xb}$	reduced heating power heat pump (first type; version $\bullet$) due to by-pass
$\dot{Q}_C(k+i)$	total cooling power of the HVAC at $k+i$
$\dot{Q}_\bullet$	cooling/heating power of component $\bullet$
$\dot{Q}_H(k+i)$	total heating power of the HVAC at $k+i$
$\boldsymbol{R}(k+i)$	plant MPC weighting matrix control
r_0	enthalpy of evaporation
$\boldsymbol{S}(k+i)$	plant MPC weighting matrix control increments
$T_{\bullet,ON}$	min. consecutive on time component $\bullet$
$T_{\bullet,OFF}$	min. consecutive off time component $\bullet$
$\boldsymbol{T}_{s,L}(k+i)$	plant MPC weighting matrix lower slack variables
$\boldsymbol{T}_{s,U}(k+i)$	plant MPC weighting matrix upper slack variables
$\boldsymbol{U}(k)$	plant MPC optimized control trajectories
$\boldsymbol{u}(k+i)$	plant MPC optimized control at $k+i$
u_j	input j plant MPC plant model
$u_{j,MILP}$	plant input j computed by MILP
$\boldsymbol{U}_{max}(k)$	time-variant upper bound on $\boldsymbol{U}(k)$
$\boldsymbol{u}_{max}(k+i)$	upper bound on control $\boldsymbol{u}(k+i)$ plant MPC
$\boldsymbol{U}_{min}(k)$	time-variant lower bound on $\boldsymbol{U}(k)$
$\boldsymbol{u}_{min}(k+i)$	lower bound on control $\boldsymbol{u}(k+i)$ plant MPC
$v_\bullet$	valve position of RM/HP (second type) $\bullet$
$\dot{W}_{t,SAF}$	power introduced by the supply air fan
$\underline{W}_\bullet(k+i)$	weight sequence for MILP objective $\bullet$
$x_{H_2O,A}$	water content wet air mass/mass flow A
$\boldsymbol{X}_{pre}(k)$	predicted state trajectories at k
$\boldsymbol{x}_{pre}(k+i\|k)$	plant MPC state predictions for $k+i$ at k
$\boldsymbol{y}(k)$	plant output (state of the indoor air)
$\boldsymbol{Y}(k)$	predicted plant output trajectories at k
$\boldsymbol{y}(k+i\|k)$	predicted plant output for $k+i$ at k
$\boldsymbol{Y}_{max}(k)$	time variant upper limit on $\boldsymbol{Y}(k)$
$\boldsymbol{Y}_{min}(k)$	time variant lower limit on $\boldsymbol{Y}(k)$
$\boldsymbol{Y}_{ref}(k)$	predicted reference trajectories for $\boldsymbol{y}$ at k
$\boldsymbol{y}_{ref}(k+i\|k)$	reference trajectories for $k+i$ at k
$\boldsymbol{Y}_{ref,MILP}(k)$	reference trajectories for MILP-MPC at k
$\boldsymbol{Y}_{s,L}(k)$	slack variables subtracted from $\boldsymbol{Y}_{min}(k)$
$\boldsymbol{Y}_{s,U}(k)$	slack variables added to $\boldsymbol{Y}_{max}(k)$
$\boldsymbol{Z}_{pre}(k)$	predicted disturbance trajectories at k
$\boldsymbol{z}_{pre}(k+i\|k)$	disturbance predictions for $k+i$ at k

Greek letters

α	fresh air ratio
$\delta_{\bullet,j}$	j-th binary variable for modeling $\bullet$
$\delta_{\bullet,\uparrow^{ON}_{OFF}}$	binary indicating if component $\bullet$ is switched on
ϑ_A	temperature of air mass/mass flow A
$\vartheta_{AC\bullet}$	temperature of the evaporator heat exchanger (RM of type & version $\bullet$)
$\lambda_{\bullet,j}$	SOS-2 variable for approximating nonlinearities of component $\bullet$
ω	rotational speed of compressor
$\Delta_\bullet$	variable for penalizing deviations/violations of $\bullet$

Subscripts

ACD	RM (first type) with discrete compressor power P_{ACD}
ACS	RM (first type) with semi-continuous compressor power P_{ACS}
ACV	RM (second type) with semi-continuous (electronic expansion/by-pass) valve
c	multiplicity counter of components with same MILP formulation
C	cooling
COND	condensed water
H	heating
HB	heater battery
HP	heat pump
HPD	HP (first type) with discrete compressor power P_{ACD}
HPS	HP (first type) with semi-continuous compressor power P_{ACS}
HPV	HP (second type) with semi-continuous (electronic expansion/by-pass) valve
IDA	indoor air
j	index counter (e.g. $j = 1,\dots,n$)
ODA	outdoor air
RCA	recirculation air
SAF	supply air fan
SUP	supply air

1. Introduction

Heating, ventilation, and air conditioning (HVAC) systems have to provide the required heating/cooling power for a plant and ensure the quality of the indoor air in terms of temperature, CO_2 level, and humidity with growing expectations of occupants. In doing so, they should be operated as energy efficient as possible. Furthermore, life expectancy of the HVAC components should be maximized. With many HVAC subsystems and consequently degrees of freedom (DOF) this becomes a challenging task. Continuous, discrete, and switched components render the overall system hybrid. Both technical and comfort-related criteria impose needs and restrictions on the HVAC operation. Dwell times of certain components (e.g compressors) have to be observed, evaporator icing must be prevented, the temperature of the supply air has to lie within a comfort zone, and many more restrictions apply.

The task is often accomplished by a rule based control strategy with well-defined switching transitions between constant operating modes. However, with several different or multiple HVAC components the number of distinct operating modes increases rapidly. Because the number of required rules grows exponentially with the number of components, the task quickly becomes impossible to solve. Furthermore, unexpected operating conditions and unsuitable switching transitions can yield trapped operating modes.

In this work a highly versatile and numerically tractable hierarchical MIMO MPC framework is proposed to meet the aforementioned requirements (see Fig. 1). On the upper-level a computationally inexpensive plant MPC enables large horizons to take care of the slow dynamics of a plant. It can effectively incorporate available disturbance forecasts if available. The plant MPC can be based on an almost arbitrary plant model. On the lower-level a MILP-MPC based on a flexible and modular MILP model which can represent a variety of different HVAC systems optimizes the HVAC operation. The MILP-MPC considers the control trajectory of the plant MPC as its reference trajectory. It minimizes power consumption and switching. Thereby it obeys the technological limitations and solves a unit commitment problem. Because the control variables of the MILP-MPC are those mixed-integer decision variables of the MILP model defining the continuous and discrete states of the HVAC components no heuristics have to be applied to account for discrete or switching HVAC components. Due to its prediction horizon the MILP-MPC can oversee the dwell times of HVAC components achieving optimized switching times. The mixed-integer linear formulation of the HVAC ensures a (near) global optimality and reduces the computational burden. Nonlinear component characteristics and the Mollier diagram are piecewise linear approximated to balance modeling accuracy with computation time.

A standard approach to solve the HVAC operation task is to apply a hierarchical control scheme. Often the continuous-valued control signal of a proportional integral derivative (PID) controller on the upper level is followed by heuristics or pulse width modulation (PWM) to account for discrete states of HVAC components on the lower level. In [1] a novel optimization algorithm based on epsilon constraint radial basis function (RBF) neural network for a self tuning PID controller of a decoupled HVAC system is presented. However, the system model of the HVAC does not contain integer or discrete variables. In [2] a fuzzy self-tuning PID controller is designed for temperature control of a vehicle climate chamber. Heuristics are applied in the lower level to account for switching HVAC equipment but yield suboptimal energy consumption. Furthermore, frequent switching increases wear and tear.

Increasing availability of accurate disturbance forecasts (like ambient temperature, occupancy, driving speed, etc.) motivates the usage of predictive controllers in the upper-level which can attain extra performance by utilizing this information.

In [3] a hierarchical control structure, including model predictive controllers (MPCs), is applied on a transport refrigeration system with a thermal energy storage (TES) unit parallel to the evaporator. The predictive controller on the upper level solves a charge scheduling problem for the TES, regarding a predicted driving profile of the delivery truck. In the lower level an MPC optimizes the compressor speed. With the proposed hierarchical optimal control approach, capturing the traffic pattern, performance is improved. However, integer or discrete decision variables are again not addressed.

The authors in [4] design a discrete MPC for an automotive air-conditioning system with a three-speed compressor. Three continuous MPCs are solved simultaneously, each associated to one compressor speed with two continuous inputs (evaporator and condenser fan frequency). The three cost values of the MPCs are compared. The three inputs corresponding to that MPC with the lowest cost value are applied. For a system with only one discrete component and only with a few discrete states this method is appropriate. However, for a whole HVAC with many switching and discrete states the computational burden of this method will increase in an exponential way. Furthermore, [4] did not yet include a power consumption model (e.g. via the COP map) in the objective function. Instead only control efforts were weighted. In this work the MILP-MPC (controlling the HVAC operation) considers static nonlinear power consumption models of all HVAC components (refrigeration machines, heat pumps, supply air fan, etc.) via piecewise linear approximation.

From existing literature a hierarchical control scheme seems advantageous to achieve computational tractability, especially if predictive controllers with large prediction horizons are applied in the upper level. However, in the lower level mixed-integer optimization should be applied. The benefit of mixed-integer optimization is that the mixed-integer nature of the decision variables (defining the HVAC operation mode) is optimally addressed, and the subsequent application of heuristics or PWM is not required. In this work the MILP-MPC of the MPC + MILP-MPC framework is based on a flexible and modular mixed-integer linear programming (MILP) model of an HVAC. Early works on mixed-integer predictive control of a hybrid system (with continuous and integer decision variables) using MILP are given by [5,6].

MILP models have been successfully applied to many thermal optimization problems. In [7] a MILP model of a distributed energy supply network integrated with electricity and hot water interchanges is presented. Ref. [8] proposes a stochastic model for steam and power system design composed of both system configuration and operating scheduling. The system model is formulated as MILP. In [9] a mathematical programming formulation for the design of water and heat exchanger networks (HENs) based on a two-step methodology is presented. In the first step, an MILP formulation is used to solve the water and energy allocation problem. In [10] a MILP optimization model for the retrofit of large scale HENs with intensified heat transfer techniques is presented. In [11] a MILP model for sizing a combined heat and power (CHP) system to be installed within a micro-grid is proposed. The objectives are minimizing the total operational cost of the micro-grid and finding the optimal size of the CHP.

All the afore-mentioned MILP models consider flows of hot/cold water, steam, and electricity, respectively. In the presented work the MILP model of the HVAC considers the required net enthalpy flows of humid air, net mass flows of CO_2, and net mass flows of water for the plant. Thereby it minimizes power consumption and wear of the HVAC.

In [12] a power grid simulation model for long term operation planning is presented. The work utilizes a MILP formulation and additionally addresses the unit commitment problem, which limits the on/off states of units. Because the MILP-model of the presented framework utilizes a prediction horizon it is capable to consider future demand in its decision upon the optimal on/off states of switched HVAC components with dwell times.

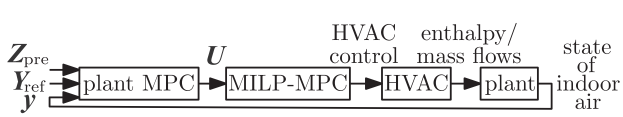


Fig. 1. Basic control architecture MPC + MILP-MPC framework.

The idea to split a complex nonlinear optimization problem into a sequential approach is computationally beneficial. In [13] a sequential approach for the synthesis of multi-period HENs is presented. An approach was proposed to solve the problem, in three sequential steps, represented by LP, MILP and NLP models. In this work also a sequential approach is chosen to solve the HVAC operation problem. First, for the plant MPC a constrained quadratic program (QP), and than for the MILP-MPC a MILP model of the HVAC is solved.

In contrast to nonlinear programming (NLP) and mixed-integer nonlinear programming (MINLP) (e.g. [14]), MILP offers the clear advantage that exact algorithms of MILP obtain the global optimum or in case of early stopping a statement with respect to optimality of the found solution [15]. The authors in [16] present a shortcut model for energy efficient water network synthesis with single contaminant. To solve the model efficiently and guarantee the solution to be the global optimum the model is formulated as a MILP. The drawback compared to NLP and MINLP is that a close approximation of nonlinearities often requires the introduction of many binary variables. In this work nonlinear component characteristics (e.g. characteristic diagram of a refrigerating machine) and the Mollier diagram are approximated as piecewise linear (with adjustable precision) to balance modeling accuracy with computation time. The MILP formulation of the HVAC ensures a (near) global optimality.

The remainder of this paper is structured as follows: In Section 2 the hierarchical MPC + MILP-MPC structure and its connection to the HVAC and plant is explained. In Section 3 the HVAC model is stated. Section 4 gives insights into the plant MPC. In Sections 5 and 6 the flexible and modular MILP optimization model of the MILP-MPC is presented. Section 7 is devoted to case studies. Section 8 gives a conclusion.

2. Hierarchical MPC + MILP-MPC structure

Fig. 2 shows the control concept of the hierarchical MPC + MILP-MPC structure.

The dynamics of the plant are slow. To optimally take available disturbance predictions into account large prediction horizons are favored. With large prediction horizons mixed-integer MIMO predictive control optimization becomes a numerically expensive problem. Therefore, the optimization problem is separated into two different controllers:

The plant MPC incorporates a dynamic model of the plant. At time sample k it obtains disturbance predictions $\boldsymbol{Z}_{\mathrm{pre}}(k) = [\boldsymbol{z}^{\mathrm{T}}_{\mathrm{pre}}(k+1|k), \ldots, \boldsymbol{z}^{\mathrm{T}}_{\mathrm{pre}}(k+N_{\mathrm{p}}|k)]^{\mathrm{T}}$, reference trajectories $\boldsymbol{Y}_{\mathrm{ref}}(k) = [\boldsymbol{y}^{\mathrm{T}}_{\mathrm{ref}}(k+1|k), \ldots, \boldsymbol{y}^{\mathrm{T}}_{\mathrm{ref}}(k+N_{\mathrm{p}}|k)]^{\mathrm{T}}$, the vectors $\boldsymbol{U}_{\min}(k) = [\boldsymbol{u}^{\mathrm{T}}_{\min}(k), \ldots, \boldsymbol{u}^{\mathrm{T}}_{\min}(k+N_{\mathrm{c}}-1)]^{\mathrm{T}}$ & $\boldsymbol{U}_{\max}(k) = [\boldsymbol{u}^{\mathrm{T}}_{\max}(k), \ldots, \boldsymbol{u}^{\mathrm{T}}_{\max}(k+N_{\mathrm{c}}-1)]^{\mathrm{T}}$ being the time-variant bounds on its optimized control trajectory (required net enthalpy and mass flows) $\boldsymbol{U}(k) = [\boldsymbol{u}^{\mathrm{T}}(k), \ldots, \boldsymbol{u}^{\mathrm{T}}(k+N_{\mathrm{c}}-1)]^{\mathrm{T}}$, and the output $\boldsymbol{y}(k)$ of the plant (state of the indoor air). The outputs of the plant MPC are the optimized control trajectory $\boldsymbol{U}(k)$ and the state predictions of its plant model $\boldsymbol{X}_{\mathrm{pre}}(k) = [\boldsymbol{x}^{\mathrm{T}}_{\mathrm{pre}}(k+1|k), \ldots, \boldsymbol{x}^{\mathrm{T}}_{\mathrm{pre}}(k+N_{\mathrm{p}}|k)]^{\mathrm{T}}$. Its control horizon N_{c} and prediction horizon N_{p} ($N_{\mathrm{c}} \le N_{\mathrm{p}}$) are chosen large enough to capture the slow thermal dynamics of the plant. The plant MPC is computationally inexpensive.

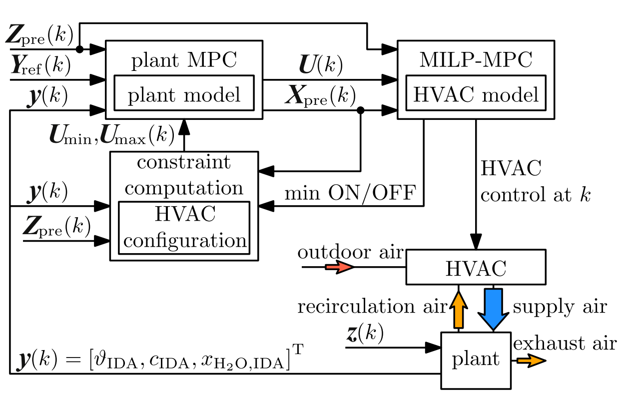


Fig. 2. Conceptual architecture of the plant MPC, the MILP-MPC, the HVAC, and the plant.

The constraint computation for the plant MPC considers the HVAC configuration (e.g. maximum refrigerating/heating capacity). It combines $\boldsymbol{y}(k)$ with the time shifted predictions $\boldsymbol{X}_{\mathrm{pre}}(k)$, gets the disturbance predictions $\boldsymbol{Z}_{\mathrm{pre}}(k)$, and combines this information to compute $\boldsymbol{U}_{\min}(k)$ and $\boldsymbol{U}_{\max}(k)$ due to technological limitations. Thereby it considers minimum consecutive ON/OFF constraints of certain HVAC components imposed by past control action of the MILP-MPC.

The MILP-MPC is based on a flexible and modular MILP optimization model of the HVAC. It obtains the disturbance predictions $\boldsymbol{Z}_{\mathrm{pre}}(k)$, the optimized control trajectory $\boldsymbol{U}(k)$ of the plant MPC and its state predictions $\boldsymbol{X}_{\mathrm{pre}}(k)$. At each time step k the MILP-MPC solves a MILP problem. It considers the first $N_{\mathrm{p,MILP}}$ samples ($N_{\mathrm{p,MILP}} \le N_{\mathrm{c}}$) of the future control trajectory $\boldsymbol{U}(k)$ of the plant MPC as its reference trajectory (i.e. $\boldsymbol{Y}_{\mathrm{ref,MILP}}(k) = [\boldsymbol{u}^{\mathrm{T}}(k), \ldots, \boldsymbol{u}^{\mathrm{T}}(k+N_{\mathrm{p,MILP}}-1)]^{\mathrm{T}}$). The control variables of the MILP-MPC are those mixed-integer decision variables of the MILP model defining the continuous and discrete states of the HVAC components over the control horizon of the MILP-MPC $N_{\mathrm{c,MILP}}$. Only the first sample $k+0$ of this sequence $k+i$ (for $i = 0, \ldots, N_{\mathrm{c,MILP}}-1$) is sent to the HVAC.

Beside tracking the reference $\boldsymbol{Y}_{\mathrm{ref,MILP}}(k)$ the objective of the MILP-MPC is to minimize power consumption and switching of the HVAC. Thereby it has to obey a variety of constraints and solve a unit commitment problem.

3. HVAC model

A modular structured HVAC model is proposed. This approach enables a quick adaption of the HVAC model to fit the structure of different HVAC system by adding or removing components. Fig. 3 shows a schematic illustration of the assumed HVAC system. Multiplicities of components are depicted aggregated.

In the mixing chamber the fresh outdoor air (ODA) mass flow $\dot{m}_{\mathrm{ODA}}$ is isobarically mixed with the recirculation air (RCA) mass flow $\dot{m}_{\mathrm{RCA}}$ from the plant interior. In Fig. 3 the quantities $h_{\mathrm{ODA}}/h_{\mathrm{IDA}}$, $x_{\mathrm{H_2O,ODA}}/x_{\mathrm{H_2O,IDA}}$, and $c_{\mathrm{ODA}}/c_{\mathrm{IDA}}$ are the specific enthalpy, the water content, and the volume fraction of CO_2 in air (in ppmv) of the wet outdoor air (ODA)/ indoor air (IDA).

In the refrigerating machines the total heat flow $\dot{Q}_{\mathrm{C}}$ is isobarically withdrawn from the wet air mass flow. If due to cooling the temperature falls below the dew point temperature the condensate is removed. In this case the leaving air is saturated air.

In heat pumps and heater batteries the total heat flows $\dot{Q}_{\mathrm{HP}}$ and $\dot{Q}_{\mathrm{HB}}$, respectively, are isobarically introduced to the wet air mass flow. The total heating power is $\dot{Q}_{\mathrm{H}}$.

The supply air fan delivers the conditioned wet air mass flow $\dot{m}_{\mathrm{SUP}}$ to the plant. Thereby it introduces the power $\dot{W}_{\mathrm{t,SAF}}$ to the air mass flow. The supply air (SUP) has the specific enthalpy h_{SUP}, water content $x_{\mathrm{H_2O,SUP}}$, and volume fraction of CO_2 in air c_{SUP}. A typical HVAC model was derived and validated in [17].

For the HVAC system depicted in Fig. 3 the system equations of the HVAC model, derived from stationary mass and energy balances, read:

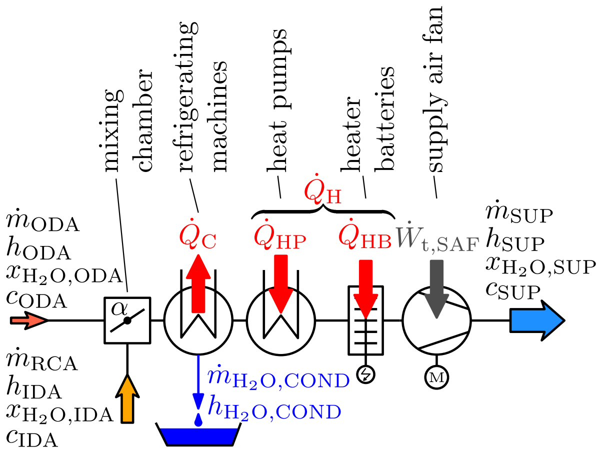


Fig. 3. Conceptual HVAC system.

$$\dot{m}_{\mathrm{SUP}}(h_{\mathrm{SUP}} - h_{\mathrm{IDA}}) = \alpha\dot{m}_{\mathrm{SUP}}(h_{\mathrm{ODA}} - h_{\mathrm{IDA}}) - \dot{Q}_{\mathrm{C}} + \dot{Q}_{\mathrm{H}} + \dot{W}_{\mathrm{t,SAF}} - \dot{m}_{\mathrm{H_2O,COND}}h_{\mathrm{H_2O,COND}} \tag{1}$$

$$\dot{m}_{\mathrm{SUP}}(c_{\mathrm{SUP}} - c_{\mathrm{IDA}}) = \alpha\dot{m}_{\mathrm{SUP}}(c_{\mathrm{ODA}} - c_{\mathrm{IDA}}) \tag{2}$$

$$\dot{m}_{\mathrm{SUP}}(x_{\mathrm{H_2O,SUP}} - x_{\mathrm{H_2O,IDA}}) = \alpha\dot{m}_{\mathrm{SUP}}(x_{\mathrm{H_2O,ODA}} - x_{\mathrm{H_2O,IDA}}) - \dot{m}_{\mathrm{H_2O,COND}} \tag{3}$$

In (1)–(3) the quantity $\alpha = \dot{m}_{\mathrm{ODA}}/\dot{m}_{\mathrm{SUP}} \leq 1$ is the fresh air ratio. It defines the state of the mixing chamber.

The HVAC model (1)–(3) is a static model. The behavior of the vapor compression plants (VCPs) is described by their coefficient of performance (COP). The COPs are given via characteristic diagrams specified by static lookup tables. A detailed description of modeling and control of VCPs can be found in [18].

4. Plant MPC

In this Section first the input to the plant model and the optimization criterion of the plant MPC are stated. Then the time-variant input constraints of the plant MPC, imposed by the hierarchical MPC + MILP-MPC structure, are derived.

4.1. MPC plant model

The output of the plant model is the temperature ϑ_{IDA}, the water content $x_{\mathrm{H_2O,IDA}}$ and volume fraction of CO_2 in air c_{IDA} of its indoor air (IDA). The plant model considers the disturbances acting on the plant imposed by its environment (radiation, ambient temperature, etc.).

The input to the plant model (provided by the HVAC) reads

$$\boldsymbol{u}(k) = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} \dot{m}_{\mathrm{SUP}}(h_{\mathrm{SUP}} - h_{\mathrm{IDA}}) \\ \dot{m}_{\mathrm{SUP}}(c_{\mathrm{SUP}} - c_{\mathrm{IDA}}) \\ \dot{m}_{\mathrm{SUP}}(x_{\mathrm{H_2O,SUP}} - x_{\mathrm{H_2O,IDA}}) \end{bmatrix}. \tag{4}$$

In (4) it is implicitly assumed that the exhaust air mass flow blown into the environment (see Fig. 2) equals the fresh air mass flow $\dot{m}_{\mathrm{ODA}}$ sucked in by the HVAC system. The exhaust air has the state of the indoor air.

Note that in (4) the right hand sides of the equations (1)–(3) are being stacked. This defines the interrelation between the decision variables of the plant MPC and those of the MILP-MPC for the HVAC operation.

A general time-variant continuous state space representation of the plant model reads

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{z}, t);\ \boldsymbol{x}(t_0) = \boldsymbol{x}_0 \tag{5}$$

$$\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{z}, t). \tag{6}$$

The time-variant vector functions $\boldsymbol{f}$ and $\boldsymbol{g}$ map the current state $\boldsymbol{x}(t)$, the input $\boldsymbol{u}(t)$, and the disturbances $\boldsymbol{z}(t)$ onto the state-derivative $\dot{\boldsymbol{x}}$ and the output $\boldsymbol{y}$, respectively. In (5) $\boldsymbol{x}_0$ denotes the initial state vector at the initial time t_0. The plant MPC is based on the zero-order hold (see [19,20]) discretized plant model.

Remark: A suitable plant model for a rail vehicle was derived and validated in [17]. Estimation of some of its parameters (i.e. heat capacity and heat transfer coefficient) based on measurements is discussed in [21]. Predictions of speed, position, and orientation of the rail vehicle can be obtained by a speed profile optimizer like the one presented in [22].

4.2. Plant MPC optimization criterion

The plant MPC optimizes the criterion

$$\begin{aligned} \min_{\boldsymbol{U},\boldsymbol{Y}_{\mathrm{s,L}},\boldsymbol{Y}_{\mathrm{s,U}}} J(k) = {} & (\boldsymbol{Y}_{\mathrm{ref}} - \boldsymbol{Y})^{\mathrm{T}}\boldsymbol{Q}(k)(\boldsymbol{Y}_{\mathrm{ref}} - \boldsymbol{Y}) \\ & + \boldsymbol{U}^{\mathrm{T}}\boldsymbol{R}(k)\boldsymbol{U} + \Delta\boldsymbol{U}^{\mathrm{T}}\boldsymbol{S}(k)\Delta\boldsymbol{U} \\ & + \boldsymbol{Y}_{\mathrm{s,L}}^{\mathrm{T}}\boldsymbol{T}_{\mathrm{s,L}}(k)\boldsymbol{Y}_{\mathrm{s,L}} \\ & + \boldsymbol{Y}_{\mathrm{s,U}}^{\mathrm{T}}\boldsymbol{T}_{\mathrm{s,U}}(k)\boldsymbol{Y}_{\mathrm{s,U}} \end{aligned} \tag{7}$$

$$\text{s.t.} \quad \boldsymbol{U}_{\min}(k) \leq \boldsymbol{U}(k) \leq \boldsymbol{U}_{\max}(k) \tag{8}$$

$$\boldsymbol{Y}_{\min}(k) - \boldsymbol{Y}_{\mathrm{s,L}}(k) \leq \boldsymbol{Y}(k) \tag{9}$$

$$\boldsymbol{Y}(k) \leq \boldsymbol{Y}_{\max}(k) + \boldsymbol{Y}_{\mathrm{s,U}}(k) \tag{10}$$

$$\boldsymbol{0} \leq \boldsymbol{Y}_{\mathrm{s,L}}(k) \tag{11}$$

$$\boldsymbol{0} \leq \boldsymbol{Y}_{\mathrm{s,U}}(k) \tag{12}$$

where $\boldsymbol{Q}$, $\boldsymbol{R}$, $\boldsymbol{S}$, $\boldsymbol{T}_{\mathrm{s,L}}$, and $\boldsymbol{T}_{\mathrm{s,U}}$ are positive semi-definite weighting matrices, which are used for tuning. The optimal control vector $\boldsymbol{U}(k)$ contains the current and future control $\boldsymbol{u}(k+i)$ (for $i = 0, \ldots, N_{\mathrm{c}} - 1$). The vector $\Delta\boldsymbol{U}(k)$ contains the control increments $\Delta\boldsymbol{u}(k+i) = \boldsymbol{u}(k+i) - \boldsymbol{u}(k+i-1)$. The vector $\boldsymbol{Y}_{\mathrm{ref}}(k)$ contains the future reference trajectory $\boldsymbol{y}_{\mathrm{ref}}(k+i|k)$ (for $i = 1, \ldots, N_{\mathrm{p}}$) at time step k. In the vector $\boldsymbol{Y}(k)$ the predictions of the output vector signal $\boldsymbol{y}(k+i|k)$ (for $i = 1, \ldots, N_{\mathrm{p}}$) at time step k are being stacked, see [23]. In the vectors $\boldsymbol{U}_{\min}(k)$ and $\boldsymbol{U}_{\max}(k)$ the time-variant constraints on the input vector $\boldsymbol{u}(k+i)$ are being stacked. The optimal parameter vectors $\boldsymbol{Y}_{\mathrm{s,L}}(k)$ and $\boldsymbol{Y}_{\mathrm{s,U}}(k)$ contain positive real slack variables subtracted from, respectively added to, the vectors $\boldsymbol{Y}_{\min}(k)$ and $\boldsymbol{Y}_{\max}(k)$ containing the lower respectively upper limit on the predicted output $\boldsymbol{y}(k+i|k)$ (i.e. (9)–(12) represent "soft" constraints on $\boldsymbol{y}(k+i|k)$).

4.3. Plant MPC time-variant input constraints

In this section the time-variant bounds $\boldsymbol{u}_{\min}(k+i)$ and $\boldsymbol{u}_{\max}(k+i)$ (for $i = 0, \ldots, N_{\mathrm{c}} - 1$) on the control input of the plant MPC (4) are derived. The bounds depend on the current and (for $i > 1$) the future state of the outdoor/indoor air, the HVAC configuration, supply air temperature constraints, and in case of minimum consecutive on/off constraints explicitly on past control action of the MILP-MPC for the HVAC.

4.3.1. Constraints imposed on u_1 due to restricted heating/cooling power

The minimum constraint on u_1 (see (4) and (1)) with respect to the enthalpy of the outdoor/indoor air $h_{\mathrm{ODA}}/h_{\mathrm{IDA}}$, the fresh air ratio α, the supply air mass flow $\dot{m}_{\mathrm{SUP}}$, the power introduced to the air mass flow by the supply air fan $\dot{W}_{\mathrm{t,SAF}}$, and the heating/cooling powers of the HVAC $\dot{Q}_{\mathrm{H}}/\dot{Q}_{\mathrm{C}}$ reads (i.e. (1) evaluated at extremes):

$$\begin{aligned} u_{1,\min}(k+i) = {} & \alpha_{\min} \cdot \dot{m}_{\mathrm{SUP,min}} \cdot (h_{\mathrm{ODA}}(k+i) - h_{\mathrm{IDA}}(k+i)) \\ & - \dot{Q}_{\mathrm{C,max}}(k+i) + \dot{Q}_{\mathrm{H,min}}(k+i) \\ & + \dot{W}_{\mathrm{t,SAF}}(\dot{m}_{\mathrm{SUP,min}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | h_{\mathrm{ODA}}(k+i) \geq h_{\mathrm{IDA}}(k+i) \end{aligned} \tag{13}$$

$$\begin{aligned} u_{1,\min}(k+i) = {} & \alpha_{\max} \dot{m}^*_{\mathrm{SUP}}(k+i)(h_{\mathrm{ODA}}(k+i) - h_{\mathrm{IDA}}(k+i)) \\ & - \dot{Q}_{\mathrm{C,max}}(k+i) + \dot{Q}_{\mathrm{H,min}}(k+i) \\ & + \dot{W}_{\mathrm{t,SAF}}(\dot{m}^*_{\mathrm{SUP}}(k+i)) \\ & \forall i \in \{0, \ldots, N_c - 1\} | h_{\mathrm{ODA}}(k+i) < h_{\mathrm{IDA}}(k+i) \end{aligned} \tag{14}$$

where $\dot{m}^*_{\mathrm{SUP}}(k+i)$ is given by

$$\dot{m}^*_{\mathrm{SUP}}(k+i) = \arg\min_{\dot{m}_{\mathrm{SUP}}} J_{u_{\min}}(\dot{m}_{\mathrm{SUP}}, k, i) \tag{15}$$

$$\begin{aligned} J_{u_{\min}}(\dot{m}_{\mathrm{SUP}}, k, i) = {} & \alpha_{\max} \dot{m}_{\mathrm{SUP}}(h_{\mathrm{ODA}}(k+i) - h_{\mathrm{IDA}}(k+i)) \\ & + \dot{W}_{\mathrm{t,SAF}}(\dot{m}_{\mathrm{SUP}}) \end{aligned} \tag{16}$$

$$\dot{m}_{\mathrm{SUP}} \in \{\dot{m}_{\mathrm{SUP,min}}, \ldots, \cdot \dot{m}_{\mathrm{SUP,max}}\} \tag{17}$$

The maximum constraints on u_1 reads:

$$\begin{aligned} u_{1,\max}(k+i) = {} & \alpha_{\max} \cdot \dot{m}_{\mathrm{SUP,max}} \cdot (h_{\mathrm{ODA}}(k+i) - h_{\mathrm{IDA}}(k+i)) \\ & - \dot{Q}_{\mathrm{C,min}}(k+i) + \dot{Q}_{\mathrm{H,max}}(k+i) \\ & + \dot{W}_{\mathrm{t,SAF}}(\dot{m}_{\mathrm{SUP,max}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | h_{\mathrm{ODA}}(k+i) \geq h_{\mathrm{IDA}}(k+i) \end{aligned} \tag{18}$$

$$\begin{aligned} u_{1,\max}(k+i) = {} & \alpha_{\min} \dot{m}^*_{\mathrm{SUP}}(k+i)(h_{\mathrm{ODA}}(k+i) - h_{\mathrm{IDA}}(k+i)) \\ & - \dot{Q}_{\mathrm{C,min}}(k+i) + \dot{Q}_{\mathrm{H,max}}(k+i) \\ & + \dot{W}_{\mathrm{t,SAF}}(\dot{m}^*_{\mathrm{SUP}}(k+i)) \\ & \forall i \in \{0, \ldots, N_c - 1\} | h_{\mathrm{ODA}}(k+i) < h_{\mathrm{IDA}}(k+i) \end{aligned} \tag{19}$$

where $\dot{m}^*_{\mathrm{SUP}}(k+i)$ in this case is given by

$$\dot{m}^*_{\mathrm{SUP}}(k+i) = \arg\max_{\dot{m}_{\mathrm{SUP}}} J_{u_{\max}}(\dot{m}_{\mathrm{SUP}}, k, i) \tag{20}$$

$$\begin{aligned} J_{u_{\max}}(\dot{m}_{\mathrm{SUP}}, k, i) = {} & \alpha_{\min} \dot{m}_{\mathrm{SUP}}(h_{\mathrm{ODA}}(k+i) - h_{\mathrm{IDA}}(k+i)) \\ & + \dot{W}_{\mathrm{t,SAF}}(\dot{m}_{\mathrm{SUP}}) \end{aligned} \tag{21}$$

$$\dot{m}_{\mathrm{SUP}} \in \{\dot{m}_{\mathrm{SUP,min}}, \ldots, \dot{m}_{\mathrm{SUP,max}}\} \tag{22}$$

It is assumed that the function $\dot{m}_{\mathrm{SUP}} = \dot{m}_{\mathrm{SUP}}(P_{\mathrm{SAF}})$ is monotonically increasing and the electric consumption of the supply air fan (SAF) adds completely to the heat input, i.e. $\dot{W}_{\mathrm{t,SAF}} = P_{\mathrm{SAF}}$.

In case of minimum consecutive on and off constraints on a refrigeration machine's compressor the refrigeration machines minimum and maximum cooling power, respectively, can be affected. In case a refrigeration machine has to stay on at some time step $k+i$ its provided minimum cooling power is different from zero. Whereas in case it can not be turned on its maximum cooling power is zero.

4.3.2. Constraints imposed on u_1 due to supply air temperature constraints

The supply air temperature ϑ_{SUP} has to lie between an upper $\vartheta_{\mathrm{SUP,max}}$ and a lower bound $\vartheta_{\mathrm{SUP,min}}$ to fulfill comfort or quality requirements. The bounds on u_1 due to this constraints are derived in the following.

From (3) the water content of the supply air is

$$x_{\mathrm{H_2O,SUP}} = \alpha x_{\mathrm{H_2O,ODA}} + (1-\alpha) x_{\mathrm{H_2O,IDA}} - \frac{\dot{m}_{\mathrm{H_2O,COND}}}{\dot{m}_{\mathrm{SUP}}}. \tag{23}$$

Neglecting the water mass flow of condensed water (i.e. $\dot{m}_{\mathrm{H_2O,COND}} \approx 0$) removed in the refrigeration machines for the moment, $x_{\mathrm{H_2O,SUP}}$ with minimum and maximum fresh air ratio α is given by:

$$x_{\mathrm{H_2O,SUP},\alpha_{\min}} = \alpha_{\min} x_{\mathrm{H_2O,ODA}} + (1-\alpha_{\min}) x_{\mathrm{H_2O,IDA}} \tag{24}$$

$$x_{\mathrm{H_2O,SUP},\alpha_{\max}} = \alpha_{\max} x_{\mathrm{H_2O,ODA}} + (1-\alpha_{\max}) x_{\mathrm{H_2O,IDA}} \tag{25}$$

The minimum and maximum water content of the supply air is then given by:

$$x^*_{\mathrm{H_2O,SUP,min}} = \min(\{x_{\mathrm{H_2O,SUP},\alpha_{\min}}, x_{\mathrm{H_2O,SUP},\alpha_{\max}}\}) \tag{26}$$

$$x_{\mathrm{H_2O,SUP,max}} = \max(\{x_{\mathrm{H_2O,SUP},\alpha_{\min}}, x_{\mathrm{H_2O,SUP},\alpha_{\max}}\}) \tag{27}$$

If $x^*_{\mathrm{H_2O,SUP,min}} > x_{\mathrm{H_2O,SUP,s}}(\vartheta_{\mathrm{SUP,min}})$, with $x_{\mathrm{H_2O,SUP,s}}$ being the saturation water content of air for $\vartheta_{\mathrm{SUP,min}}$ (for a given air pressure), water is condensed in the refrigeration machines and the condensed water is removed. Thus the maximum water content of the supply air with temperature $\vartheta_{\mathrm{SUP,min}}$ is given by

$$x_{\mathrm{H_2O,SUP,min}} = \min(\{x^*_{\mathrm{H_2O,SUP,min}}, x_{\mathrm{H_2O,SUP,s}}(\vartheta_{\mathrm{SUP,min}})\}) \tag{28}$$

and $\dot{m}_{\mathrm{H_2O,COND}}/\dot{m}_{\mathrm{SUP}} = x^*_{\mathrm{H_2O,SUP,min}} - x_{\mathrm{H_2O,SUP,min}}$. Thus the minimum and maximum enthalpy of the supply air are:

$$h_{\mathrm{SUP,min}} = c_{\mathrm{p,a}} \vartheta_{\mathrm{SUP,min}} + x_{\mathrm{H_2O,SUP,min}}(c_{\mathrm{p,v}} \vartheta_{\mathrm{SUP,min}} + r_0) \tag{29}$$

$$h_{\mathrm{SUP,max}} = c_{\mathrm{p,a}} \vartheta_{\mathrm{SUP,max}} + x_{\mathrm{H_2O,SUP,max}}(c_{\mathrm{p,v}} \vartheta_{\mathrm{SUP,max}} + r_0) \tag{30}$$

The enthalpy of evaporation r_0, the specific heat capacities of dry air $c_{\mathrm{p,a}}$ and water vapor $c_{\mathrm{p,v}}$ are considered constant. The lower and upper bound on $u_1(k+i)$ due to supply air temperature constraints read:

$$u_{1,\min}(k+i) = \dot{m}_{\mathrm{SUP,max}}(h_{\mathrm{SUP,min}}(k+i) - h_{\mathrm{IDA}}(k+i)) \tag{31}$$

$$u_{1,\max}(k+i) = \dot{m}_{\mathrm{SUP,max}}(h_{\mathrm{SUP,max}}(k+i) - h_{\mathrm{IDA}}(k+i)) \tag{32}$$

In (31) and (32) it is assumed that $h_{\mathrm{SUP,min}}(k+i) \leq h_{\mathrm{IDA}}(k+i) \leq h_{\mathrm{SUP,max}}(k+i)$ holds.

The current plant output and the state predictions of the previous time step $k-1$ are utilized to formulate the constraints at the current time step k over the prediction horizon $k+i$ (for $i = 0, \ldots, N_c - 1$).

4.3.3. Constraints imposed on u_2

The minimum constraint on u_2 (see (4) and (2)) reads:

$$\begin{aligned} & u_{2,\min}(k+i) = \alpha_{\min} \dot{m}_{\mathrm{SUP,min}}(c_{\mathrm{ODA}} - c_{\mathrm{IDA}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | c_{\mathrm{ODA}}(k+i) \geq c_{\mathrm{IDA}}(k+i) \end{aligned} \tag{33}$$

$$\begin{aligned} & u_{2,\min}(k+i) = \alpha_{\max} \dot{m}_{\mathrm{SUP,max}}(c_{\mathrm{ODA}} - c_{\mathrm{IDA}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | c_{\mathrm{ODA}}(k+i) < c_{\mathrm{IDA}}(k+i) \end{aligned} \tag{34}$$

Maximum constraints on u_2 are not imposed since the introduction of CO_2 to the plant is not an objective.

4.3.4. Constraints imposed on u_3

The minimum constraint on u_3 (see (4) and (3)) reads:

$$\begin{aligned} & u_{3,\min}(k+i) = \dot{m}_{\mathrm{SUP,min}}(x_{\mathrm{H_2O,SUP,min}} - x_{\mathrm{H_2O,IDA}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | x_{\mathrm{H_2O,SUP,min}}(k+i) \geq x_{\mathrm{H_2O,IDA}}(k+i) \end{aligned} \tag{35}$$

$$\begin{aligned} & u_{3,\min}(k+i) = \dot{m}_{\mathrm{SUP,max}}(x_{\mathrm{H_2O,SUP,min}} - x_{\mathrm{H_2O,IDA}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | x_{\mathrm{H_2O,SUP,min}}(k+i) < x_{\mathrm{H_2O,IDA}}(k+i) \end{aligned} \tag{36}$$

with $x_{\mathrm{H_2O,SUP,min}}$ from (28). Eq. (35) and (36) assume that the cooling power of the HVAC is large enough to achieve $\vartheta_{\mathrm{SUP,min}}$ independent of the enthalpy of the outdoor/indoor air and $\dot{m}_{\mathrm{SUP}}$.

The maximum constraint on u_3 reads:

$$\begin{aligned} & u_{3,\max}(k+i) = \alpha_{\max} \dot{m}_{\mathrm{SUP,max}}(x_{\mathrm{H_2O,ODA}} - x_{\mathrm{H_2O,IDA}}) \\ & \forall i \in \{0, \ldots, N_c - 1\} | x_{\mathrm{H_2O,ODA}}(k+i) \geq x_{\mathrm{H_2O,IDA}}(k+i) \end{aligned} \tag{37}$$

$$u_{3,\max}(k+i) = \alpha_{\min}\dot{m}_{\mathrm{SUP,min}}(x_{\mathrm{H_2O,ODA}} - x_{\mathrm{H_2O,IDA}})$$
$$\forall i \in \{0,\dots,N_c-1\}|x_{\mathrm{H_2O,ODA}}(k+i) < x_{\mathrm{H_2O,IDA}}(k+i) \quad (38)$$

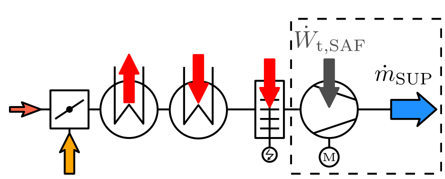
5. MILP optimization model of the HVAC: MILP formulation of the HVAC components

In this section the modular MILP formulation for the HVAC components is presented; it can handle a multiplicity of components. Beside the mixed-integer decision variables defining the state of the HVAC components the MILP model contains additional auxiliary decision variables. These continuous or binary variables are utilized for modeling product terms of decision variables, identifying switching, and piecewise linear approximation of nonlinear mappings (e.g COP maps, the Mollier diagram, etc.).

5.1. Notation of the MILP optimization model

To simplify notation the time instance $k+i$ is omitted whenever possible, i.e. when the terms do not contain mixed time instances like $k+i$ and $k+j$ with $i \neq j$. Numerical values provided to the MILP optimizer are marked by an underline, i.e. $\underline{\alpha}_{\min}$, $\underline{\alpha}_j$, $\underline{\alpha}_{\max}$ are numeric values the decision variable α of the MILP model can take. Specific quantities like the enthalpy of the indoor air $\underline{h}_{\mathrm{IDA}}(k+i)$ are computed based on the state predictions of the plant MPC and are constants to the MILP model, updated at each time step k. The subscript "c" is utilized to count multiplicity of components modeled by the same MILP formulation.

5.2. Supply air fan (SAF)



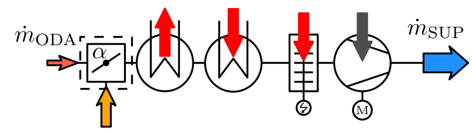
If the HVAC is in operation the supply air fan is on. I.e. the supply air mass flow $\dot{m}_{\mathrm{SUP}}$, the electrical power consumption of the SAF P_{SAF}, and the power $\dot{W}_{\mathrm{t,SAF}}$ introduced to the air mass flow are larger than zero.

Variable speed SAF: The power consumption of the SAF P_{SAF} is modeled as a continuous decision variable with $\underline{P}_{\mathrm{SAF,min}} \leq P_{\mathrm{SAF}} \leq \underline{P}_{\mathrm{SAF,max}}$. The nonlinear relationship between $\dot{m}_{\mathrm{SUP}}$ and P_{SAF} is approximated by introducing n_{SAF} special ordered set of type 2 (SOS-2) variables $\lambda_{\mathrm{SAF},j}$ (see [24]). By claiming that at most two $\lambda_{\mathrm{SAF},j}$ are non-zero and smaller 1, the non-zero variables are consecutive in their ordering, and $\sum \lambda_{\mathrm{SAF},j} = 1$ holds, they enable a piecewise linear approximation.

Discrete speed SAF: The power consumption of the SAF P_{SAF} is modeled as a discrete decision variable i.e. $P_{\mathrm{SAF}} \in \{\underline{P}_{\mathrm{SAF,min}},\dots,\underline{P}_{\mathrm{SAF},j},\dots,\underline{P}_{\mathrm{SAF,max}}\}$ and $\dot{m}_{\mathrm{SUP}} \in \{\underline{\dot{m}}_{\mathrm{SUP,min}},\dots,\underline{\dot{m}}_{\mathrm{SUP},j},\dots,\underline{\dot{m}}_{\mathrm{SUP,max}}\}$. The quantities P_{SAF} and $\dot{m}_{\mathrm{SUP}}$ are modeled by

$$\sum_j \delta_{\mathrm{SAF},j} = 1 \qquad \delta_{\mathrm{SAF},j} \in \{0,1\}\,\forall j$$
$$P_{\mathrm{SAF}} = \sum_j \underline{P}_{\mathrm{SAF},j}\cdot\delta_{\mathrm{SAF},j} \quad (39)$$
$$\dot{m}_{\mathrm{SUP}} = \sum_j \underline{\dot{m}}_{\mathrm{SUP},j}\cdot\delta_{\mathrm{SAF},j}$$

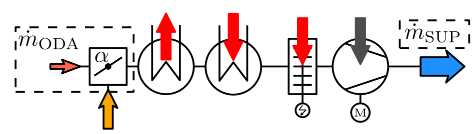
5.3. Fresh air ratio α



The discrete decision variable $\alpha = \dot{m}_{\mathrm{ODA}}/\dot{m}_{\mathrm{SUP}} \in \{\underline{\alpha}_{\min},\dots,\underline{\alpha}_j,\dots,\underline{\alpha}_{\max}\}$ (with $\underline{\alpha}_{\min} \geq 0$ and $\underline{\alpha}_{\max} \leq 1$) defines the flap/rotary slide position in the mixing chamber and is modeled as follows:

$$\alpha = \sum_j \underline{\alpha}_j \cdot \delta_{\alpha,j} \qquad \delta_{\alpha,j} \in \{0,1\}\,\forall j \qquad \sum_j \delta_{\alpha,j} = 1 \quad (40)$$

5.4. Fresh air mass flow $\dot{m}_{\mathrm{ODA}}$ as auxiliary decision variable



The modeling of the product term $\dot{m}_{\mathrm{SUP}}\cdot\alpha$ in (1)–(3) is achieved by introducing the fresh air mass flow $\dot{m}_{\mathrm{ODA}} = \dot{m}_{\mathrm{SUP}}\cdot\alpha$ as auxiliary decision variable. For $\dot{m}_{\mathrm{ODA}}$ the following box constraint has to hold:

$$\underline{\dot{m}}_{\mathrm{SUP,min}}\underline{\alpha}_{\min} \leq \dot{m}_{\mathrm{ODA}} \leq \underline{\dot{m}}_{\mathrm{SUP,max}}\underline{\alpha}_{\max} \quad (41)$$

The auxiliary variable $\dot{m}_{\mathrm{ODA}}$ (continuous variable) is connected to the fresh air mass flow $\dot{m}_{\mathrm{SUP}}$ (continuous variable) and the fresh air ratio α (discrete variable) by the following two inequality constraints:

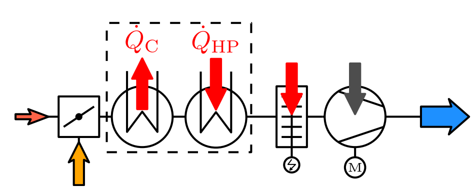
$$m_{\alpha_j} \leq \dot{m}_{\mathrm{ODA}} - \dot{m}_{\mathrm{SUP}}\cdot\underline{\alpha}_j \leq M_{\alpha_j} \quad (42)$$

with

$$m_{\alpha_j} = (\underline{\dot{m}}_{\mathrm{SUP,min}}\underline{\alpha}_{\min} - \underline{\dot{m}}_{\mathrm{SUP,max}}\underline{\alpha}_j)\cdot(1-\delta_{\alpha,j}) \quad (43)$$
$$M_{\alpha_j} = (\underline{\dot{m}}_{\mathrm{SUP,max}}\underline{\alpha}_{\max} - \underline{\dot{m}}_{\mathrm{SUP,min}}\underline{\alpha}_j)\cdot(1-\delta_{\alpha,j}) \quad (44)$$

Note that if $\delta_{\alpha,j} = 1$ the decision variable $\alpha = \underline{\alpha}_j$.

5.5. Refrigeration machines and heat pumps



Only the modeling of the refrigeration machines ("AC") is presented in this section since heat pumps ("HP") are modeled identically. The presented MILP model can distinguish between two main refrigeration machine model types.

The first type is presented in Section 5.5.1. For this type the cooling power $\dot{Q}_{\mathrm{AC}}$ is modeled as a function of the compressor power P_{AC}. It can be subdivided into two versions. The first version "ACS" considers P_{AC} to be a semi-continuous decision variable (either zero or real valued), denoted by P_{ACS}. The "S" stands for semi-continuous. The second version "ACD" of the first type considers P_{AC} to be a discrete decision variable, denoted by P_{ACD}. The "D" stands for discrete. For both versions ("ACS" &"ACD") of the first type a by-pass valve can be added. The modeling of the by-pass valve is explained in the last paragraph of Section 5.5.1.

The second type ("ACV") is presented in Section 5.5.2. For the second type the cooling power $\dot{Q}_{ACV}$ and the electrical power P_{ACV} are modeled both as a function of the discrete compressor speed and the semi-continuous (electronic expansion/by-pass) valve position.

5.5.1. Cooling power as a function of the electrical compressor power and by-pass valve state

Variable speed compressors: The compressor power $P_{ACS,c}$ is modeled as a semi-continuous decision variable i.e. $P_{ACS,c} = 0 \vee \underline{P}_{ACS,c,min} \leq P_{ACS,c} \leq \underline{P}_{ACS,c,max}$. With the binary $\delta_{ACS,ON,c}$ which is 1 if the compressor is ON (otherwise 0) this is modeled by

$$\underline{P}_{ACS,c,min}\delta_{ACS,ON,c} \leq P_{ACS,c} \leq \underline{P}_{ACS,c,max}\delta_{ACS,ON,c} \tag{45}$$

The nonlinear relationship between the cooling power $\dot{Q}_{ACS,c}$ and $P_{ACS,c}$ is approximated by introducing $n_{ACS,c}$ SOS-2 variables $\lambda_{ACS,c,j}$:

$$P_{ACS,c} = \sum_{j}^{n_{ACS,c}} \underline{P}_{ACS,c,j} \cdot \lambda_{ACS,c,j} \tag{46}$$

$$\dot{Q}_{ACS,c} = \sum_{j}^{n_{ACS,c}} \underline{\dot{Q}}_{ACS,c,j} \cdot \lambda_{ACS,c,j} \tag{47}$$

with $\underline{P}_{ACS,c,1} = 0$, $\underline{P}_{ACS,c,2} = \underline{P}_{ACS,c,min}$, $\underline{P}_{ACS,c,n_{ACS,c}} = \underline{P}_{ACS,c,max}$ and $\underline{\dot{Q}}_{ACS,c,j}(k+i) = \underline{\dot{Q}}_{ACS,c,j}(\vartheta_{ODA}(k+i), \vartheta_{ACS,c}(k+i))$. The quantities $\underline{P}_{ACS,c,j}$ and $\underline{\dot{Q}}_{ACS,c,j}$ denote the supporting points of the piecewise linear approximation. To fulfill the order relation their numerical values are sorted in ascending order of $\underline{P}_{ACS,c,j}$. The quantities $\vartheta_{ODA}(k+i)$ and $\vartheta_{ACS,c}(k+i)$ are the outdoor air temperature and the temperature of the evaporator heat exchanger. For $\vartheta_{ACS,c}(k+i)$ assumptions have to be made, whereas for $\vartheta_{ODA}(k+i)$ the disturbance predictions are utilized. With $n_{ACS,c} - 1$ auxiliary binaries $\delta_{ACS,c,j}$ the following constraints have to hold (see [15]):

$$\sum_{j}^{n_{ACS,c}-1} \delta_{ACS,c,j} = 1; \text{ with } \delta_{ACS,c,j} \in \{0,1\} \forall j \tag{48}$$

$$\sum_{j}^{n_{ACS,c}} \lambda_{ACS,c,j} = 1; \text{ with } \lambda_{ACS,c,j} \in [0,1] \forall j \tag{49}$$

$$\lambda_{ACS,c,1} \leq \delta_{ACS,c,1} \tag{50}$$

$$\lambda_{ACS,c,j} \leq \delta_{ACS,c,j-1} + \delta_{ACS,c,j}; \quad j = 2, \ldots, n_{ACS,c} - 1 \tag{51}$$

$$\lambda_{ACS,c,n_{ACS,c}} \leq \delta_{ACS,c,n_{ACS,c}-1} \tag{52}$$

Discrete speed compressors: The compressor power $P_{ACD,c}$ is modeled as a discrete decision variable i.e. $P_{ACD,c} \in \{0, \ldots, \underline{P}_{ACD,c,j}, \ldots, \underline{P}_{ACD,c,max}\}$. With the binary $\delta_{ACD,c,j}$ which is 1 if $P_{ACD,c} = \underline{P}_{ACD,c,j}$ (otherwise 0) $P_{ACD,c}$ and $\dot{Q}_{ACD,c}$ are modeled by

$$P_{ACD,c} = \sum_{j}^{n_{ACD,c}} \underline{P}_{ACD,c,j} \cdot \delta_{ACD,c,j} \tag{53}$$

$$\dot{Q}_{ACD,c} = \sum_{j}^{n_{ACD,c}} \underline{\dot{Q}}_{ACD,c,j} \cdot \delta_{ACD,c,j} \tag{54}$$

$$\sum_{j}^{n_{ACD,c}} \delta_{ACD,c,j} = 1; \text{ with } \delta_{ACD,c,j} \in \{0,1\} \forall j \tag{55}$$

with $\underline{\dot{Q}}_{ACD,c,j}(k+i) = \underline{\dot{Q}}_{ACD,c,j}(\vartheta_{ODA}(k+i), \vartheta_{ACD,c}(k+i))$. Where (again) $\vartheta_{ACD,c}(k+i)$ is the assumed temperature of the evaporator heat exchanger.

Refrigeration machine with by-pass valve: The cooling power of a refrigeration machine ("ACS" and "ACD" type) with a by-pass valve is computed by

$$\dot{Q}_{ACxb,c} = \dot{Q}_{AC,c} \cdot b_{AC,c}. \tag{56}$$

In (56) the subscripts "S" and "D" of "ACS" and "ACD" where neglected. Both versions are modeled as follows.

The by-pass valve state is considered a discrete decision variable $b_{AC,c} \in \{\underline{b}_{AC,c,min}, \ldots, \underline{b}_{AC,c,j}, \ldots, \underline{b}_{AC,c,max}\}$ (with $\underline{b}_{AC,c,min} > 0$ and $\underline{b}_{AC,c,max} = 1$). It is modeled by:

$$b_{AC,c} = \sum_{j} \underline{b}_{AC,c,j} \cdot \delta_{b_{AC},c,j} \tag{57}$$

$$\sum_{j} \delta_{b_{AC},c,j} = 1; \text{ with } \delta_{b_{AC},c,j} \in \{0,1\} \forall j \tag{58}$$

If $b_{AC,c} = \underline{b}_{AC,c,max} = 1$ the by-pass valve is not used. If $b_{AC,c} < 1$ the by-pass valve is used and the cooling power of its associated refrigeration machine is reduced by $b_{AC,c}$, see (56). Modeling of the product term in (56) requires the introduction of the auxiliary decision variable $\dot{Q}_{ACxb,c}$ and connecting it to $\dot{Q}_{AC,c}$ and $b_{AC,c}$. This is done as outlined in the following.

The reduced cooling power $\dot{Q}_{ACxb,c}$ has to fulfill

$$\underline{\dot{Q}}_{AC,c,min}\underline{b}_{AC,c,min} \leq \dot{Q}_{ACxb,c} \leq \underline{\dot{Q}}_{AC,c,max}\underline{b}_{AC,c,max}. \tag{59}$$

The auxiliary variable $\dot{Q}_{ACxb,c}$ (semi-continuous or discrete variable) is connected to $\dot{Q}_{AC,c}$ and $b_{AC,c}$ by:

$$\underline{m}_{b_{AC},c,j}(1 - \delta_{b_{AC},c,j}) \leq \dot{Q}_{ACxb,c} - \dot{Q}_{AC,c}\underline{b}_{AC,c,j} \tag{60}$$

$$\dot{Q}_{ACxb,c} - \dot{Q}_{AC,c}\underline{b}_{AC,c,j} \leq \underline{M}_{b_{AC},c,j}(1 - \delta_{b_{AC},c,j}) \tag{61}$$

with

$$\underline{m}_{b_{AC},c,j} = (\underline{\dot{Q}}_{AC,c,min}\underline{b}_{AC,c,min} - \underline{\dot{Q}}_{AC,c,max}\underline{b}_{AC,c,j}) \tag{62}$$

$$\underline{M}_{b_{AC},c,j} = (\underline{\dot{Q}}_{AC,c,max}\underline{b}_{AC,c,max} - \underline{\dot{Q}}_{AC,c,min}\underline{b}_{AC,c,j}) \tag{63}$$

5.5.2. Cooling power and electrical power as a function of compressor speed and valve position

The rotational speed of the compressor $\omega_{ACV,c}$ is a discrete decision variable i.e. $\omega_{ACV,c} \in \{0, \ldots, \underline{\omega}_{ACV,c,m}, \ldots, \underline{\omega}_{ACV,c,max}\}$. With the binary $\delta_{ACV,c,m}$ which is 1 if $\omega_{ACV,c} = \underline{\omega}_{ACV,c,m}$ (otherwise 0) this is modeled by

$$\omega_{ACV,c} = \sum_{m}^{n_{ACV,c}} \underline{\omega}_{ACV,c,m} \cdot \delta_{ACV,c,m} \tag{64}$$

$$\sum_{m}^{n_{ACV,c}} \delta_{ACV,c,m} = 1; \text{ with } \delta_{ACV,c,m} \in \{0,1\} \forall m \tag{65}$$

The position of the (electronic/by-pass) valve $\upsilon_{ACV,c}$ is considered to be a semi-continuous variable i.e. $\upsilon_{ACV,c} = 0 \vee \underline{\upsilon}_{ACV,c,min} \leq \upsilon_{ACV,c} \leq \underline{\upsilon}_{ACV,c,max}$. The nonlinear mappings $P_{ACV,c} = P_{ACV,c}(\upsilon_{ACV,c}, \omega_{ACV,c})$ and $\dot{Q}_{ACV,c} = \dot{Q}_{ACV,c}(\upsilon_{ACV,c}, \omega_{ACV,c}, \vartheta_{ODA}, \vartheta_{ACV,c})$ are modeled by piecewise linear approximation as follows:

$$\lambda_{ACV,c,m,j} \in [0,1] \forall j \forall m; \qquad \delta_{ACV,c,m,j} \in \{0,1\} \forall j \forall m \tag{66}$$

$$\sum_{j}^{n_{ACV,c,m}-1} \delta_{ACV,c,m,j} = \delta_{ACV,c,m} \forall m \tag{67}$$

$$\sum_{j}^{n_{ACV,c,m}} \lambda_{ACV,c,m,j} = \delta_{ACV,c,m} \forall m \tag{68}$$

$$\lambda_{ACV,c,m,1} \leq \delta_{ACV,c,m,1} \tag{69}$$

$$\lambda_{ACV,c,m,j} \leq \delta_{ACV,c,m,j-1} + \delta_{ACV,c,m,j}; j = 2, \ldots, n_{ACV,c,m} - 1 \tag{70}$$

$$\lambda_{ACV,c,m,n_{ACV,c,m}} \leq \delta_{ACV,c,m,n_{ACV,c,m}-1} \tag{71}$$

$$\upsilon_{ACV,c,m} = \sum_{j}^{n_{ACV,c,m}} \underline{\upsilon}_{ACV,c,m,j}\lambda_{ACV,c,m,j} \tag{72}$$

with $\underline{\nu}_{\text{ACV},c,m,1} = 0$, $\underline{\nu}_{\text{ACV},c,m,2} = \underline{\nu}_{\text{ACV},c,\min}$, and $\underline{\nu}_{\text{ACV},c,m,n_{\text{ACV},c,m}} = \underline{\nu}_{\text{ACV},c,\max}$. The numerical values $\underline{\nu}_{\text{ACV},c,m,j}$ are sorted in ascending order.

$$P_{\text{ACV},c,m} = \sum_{j}^{n_{\text{ACV},c,m}} \underline{P}_{\text{ACV},c,m,j} \cdot \lambda_{\text{ACV},c,m,j} \tag{73}$$

$$\dot{Q}_{\text{ACV},c,m} = \sum_{j}^{n_{\text{ACV},c,m}} \underline{\dot{Q}}_{\text{ACV},c,m,j} \cdot \lambda_{\text{ACV},c,m,j} \tag{74}$$

with $\underline{\dot{Q}}_{\text{ACV},c,m,j}(k+i) = \underline{\dot{Q}}_{\text{ACV},c,m,j}(\vartheta_{\text{ODA}}(k+i), \vartheta_{\text{ACV},c}(k+i))$. For each m the quantities $\underline{P}_{\text{ACV},c,m,j}$ and $\underline{\dot{Q}}_{\text{ACV},c,m,j}$ denote the supporting points of the piecewise linear approximation. Their numerical values are sorted in ascending order of $\underline{\nu}_{\text{ACV},c,m,j}$.

The position of the (electronic/by-pass) valve, the cooling power, and the electrical power are obtained by

$$\nu_{\text{ACV},c} = \sum_{m}^{n_{\text{ACV},c}} \nu_{\text{ACV},c,m} \tag{75}$$

$$\dot{Q}_{\text{ACV},c} = \sum_{m}^{n_{\text{ACV},c}} \dot{Q}_{\text{ACV},c,m} \tag{76}$$

$$P_{\text{ACV},c} = \sum_{m}^{n_{\text{ACV},c}} P_{\text{ACV},c,m} \tag{77}$$

5.5.3. Minimum consecutive ON/OFF constraints for compressors

If a compressor of a refrigeration machine (type "ACS", "ACD", or "ACV") is switched on at some time instance $k+i$ it has to stay on for at least $T_{\text{AC,ON},c}$ seconds respectively $n_{\text{AC,ON},c} = T_{\text{AC,ON},c}/T_s$ time samples. The latter assumes that $T_{\text{AC,ON},c}$ is a whole-number ratio of the sampling time T_s. If at time instance $k+i$ the compressor is running the binary variable $\delta_{\text{AC,ON},c}(k+i)$ takes a value of 1 otherwise 0. The minimum consecutive on constraints for the compressor read

$$\begin{aligned}&\delta_{\text{AC,ON},c}(k+r) \geq \delta_{\text{AC,ON},c}(k+i) - \delta_{\text{AC,ON},c}(k+i-1)\\ &\forall i \in \{-n_{\text{AC,ON},c}+1, \ldots, 0, \ldots, N_{\text{c,MILP}}-1\}\\ &\forall r \in \{i, \ldots, \min(N_{\text{c,MILP}}-1, i+n_{\text{AC,ON},c}-1)\} \wedge r \geq 0.\end{aligned} \tag{78}$$

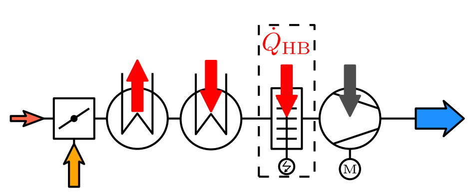
5.5.4. Refrigeration machines operable as heat pumps

If a refrigeration machine can be operated as heat pump it can not be in both modes at the same time. This is modeled by

$$\delta_{\text{AC,ON},c} \leq 1 - \delta_{\text{HP,ON},c} \tag{79}$$

$$\delta_{\text{HP,ON},c} \leq 1 - \delta_{\text{AC,ON},c}. \tag{80}$$

5.6. Heater batteries (HB)



The electrical power consumption of a heater battery $P_{\text{HB},c}$ is either modeled as a discrete, continuous, or semi-continuous variable. Its heating power equals its electrical power consumption. I.e. $\dot{Q}_{\text{HB},c} = P_{\text{HB},c}$.

5.7. Total cooling and heating power

With the different refrigeration machines/heat pumps (Section 5.5) and the heater batteries (Section 5.6) the total cooling $\dot{Q}_{\text{C}}(k+i)$ and heating power $\dot{Q}_{\text{H}}(k+i)$ at time instance $k+i$ (for $i = 0, \ldots, N_{\text{p,MILP}}-1$) read

$$\dot{Q}_{\text{C}} = \sum_{c}^{n_{\text{ACS}}} \dot{Q}_{\text{ACS},c} + \sum_{c}^{n_{\text{ACD}}} \dot{Q}_{\text{ACD},c} + \sum_{c}^{n_{\text{ACV}}} \dot{Q}_{\text{ACV},c} \tag{81}$$

$$\dot{Q}_{\text{H}} = \sum_{c}^{n_{\text{HPS}}} \dot{Q}_{\text{HPS},c} + \sum_{c}^{n_{\text{HPD}}} \dot{Q}_{\text{HPD},c} + \sum_{c}^{n_{\text{HPV}}} \dot{Q}_{\text{HPV},c} + \sum_{c}^{n_{\text{HB}}} \dot{Q}_{\text{HB},c} \tag{82}$$

6. MILP optimization of the HVAC: objectives

In this section first the overall cost function of the MILP optimization is presented. Then its individual cost and penalty functions are derived.

6.1. Overall MILP cost function at time step k

The overall cost function of the MILP at k is given by

$$\begin{aligned}J(k) = {} & J_{\text{P}}(k) + J_{\text{p,u}_1}(k) + J_{\text{p,u}_2}(k) + J_{\text{p,u}_3}(k)\\ & + J_{\text{p,u}_1,\vartheta_{\text{SUP}_{\min}\text{v}}}(k) + J_{\text{p,u}_1,\vartheta_{\text{SUP}_{\max}\text{v}}}(k)\\ & + J_{\text{p,ACS},\uparrow^{\text{ON}}_{\text{OFF}},c}(k) + J_{\text{p,ACS,change},c}(k)\\ & + \ldots\\ & + J_{\text{p,HPV},\uparrow^{\text{ON}}_{\text{OFF}},c}(k) + J_{\text{p,HPV,change},c}(k)\end{aligned} \tag{83}$$

It sums up a variety of individual cost and penalty functions. These individual contributions are outlined in the respective sections in the following.

6.2. Cost function term for power consumption

The MILP model computes the power consumption of the HVAC at time instance $k+i$ by

$$\begin{aligned}P(k+i) = {} & P_{\text{SAF}}(k+i) + \sum_{c}^{n_{\text{HB}}} P_{\text{HB},c}(k+i)\\ & + \sum_{c}^{n_{\text{ACS}}} P_{\text{ACS},c}(k+i) + \sum_{c}^{n_{\text{HPS}}} P_{\text{HPS},c}(k+i)\\ & + \sum_{c}^{n_{\text{ACD}}} P_{\text{ACD},c}(k+i) + \sum_{c}^{n_{\text{HPD}}} P_{\text{HPD},c}(k+i)\\ & + \sum_{c}^{n_{\text{ACV}}} P_{\text{ACV},c}(k+i) + \sum_{c}^{n_{\text{HPV}}} P_{\text{HPV},c}(k+i)\\ & \forall i \in \{0, \ldots, N_{\text{c}}-1\}\end{aligned} \tag{84}$$

$$\begin{aligned}P(k+i) = {} & P(k+i_{\text{f}})\\ & \text{with}\, i_{\text{f}} = N_{\text{c,MILP}}-1\\ & \forall i \in \{N_{\text{c,MILP}}, \ldots, N_{\text{p,MILP}}-1\}\end{aligned} \tag{85}$$

With the weight $\underline{W}_{\text{P}}(k+i)$ the cost function term with respect to power consumption added to the overall cost function (83) reads

$$J_{P}(k) = \sum_{i=0}^{N_{p,MILP}-1} \underline{W}_{P}(k+i) \cdot P(k+i) \tag{86}$$

6.3. Penalizing deviations from the desired net enthalpy flow

The MILP model computes (1) by

$$\begin{aligned} u_{1,MILP}(k+i) = {} & \dot{m}_{ODA}(k+i)(\underline{h}_{ODA}(k+i) - \underline{h}_{IDA}(k+i)) \\ & - \dot{Q}_{C}(k+i) + \dot{Q}_{H}(k+i) + \dot{W}_{t,SAF}(k+i) \\ & - \dot{H}_{H_2O,COND}(k+i) \\ & \forall i \in \{0,\dots,N_{c,MILP}-1\} \end{aligned} \tag{87}$$

$$\begin{aligned} u_{1,MILP}(k+i) = {} & \dot{m}_{ODA}(k+i_f)(\underline{h}_{ODA}(k+i) - \underline{h}_{IDA}(k+i)) \\ & - \dot{Q}_{C}(k+i) + \dot{Q}_{H}(k+i) + \dot{W}_{t,SAF}(k+i_f) \\ & - \dot{H}_{H_2O,COND}(k+i) \\ & \text{with}\, i_f = N_{c,MILP}-1 \\ & \forall i \in \{N_{c,MILP},\dots,N_{p,MILP}-1\} \end{aligned} \tag{88}$$

In (87) and (88) the quantity $\dot{H}_{H_2O,COND}$ is the enthalpy flow of the removed condensed water. The detailed MILP formulation for computing $\dot{H}_{H_2O,COND}$ and the mass flow of the removed condensed water $\dot{m}_{H_2O,COND}$ is not given here as it would exceed the scope of this paper.

With the auxiliary decision variable $\Delta_{u_1}(k+i) \geq 0$ the deviation between the first input of the plant MPC u_1 and the net enthalpy flow provided by the MILP HVAC $u_{1,MILP}$ is constrained by

$$\begin{aligned} -\Delta_{u_1}(k+i) \leq \underline{u}_1(k+i) - u_{1,MILP}(k+i) \leq \Delta_{u_1}(k+i) \\ \forall i \in \{0,\dots,N_{p,MILP}-1\} \end{aligned}$$

Let $\underline{W}_{u_1}(k+i)$ be a penalty weight. The penalty term penalizing deviations from the desired net enthalpy flow $\underline{u}_1(k+i)$ introduced to the plant added to (83) reads

$$J_{p,u_1}(k) = \sum_{i=0}^{N_{p,MILP}-1} \underline{W}_{u_1}(k+i) \cdot \Delta_{u_1}(k+i) \tag{89}$$

6.4. Penalizing positive deviations from the desired net mass flow of CO_2

The MILP model computes (2) by

$$\begin{aligned} u_{2,MILP}(k+i) = {} & \dot{m}_{ODA}(k+i)(\underline{c}_{ODA}(k+i) - \underline{c}_{IDA}(k+i)) \\ & \forall i \in \{0,\dots,N_{c,MILP}-1\} \end{aligned} \tag{90}$$

$$\begin{aligned} u_{2,MILP}(k+i) = {} & \dot{m}_{ODA}(k+i_f)(\underline{c}_{ODA}(k+i) - \underline{c}_{IDA}(k+i)) \\ & \text{with}\, i_f = N_{c,MILP}-1 \\ & \forall i \in \{N_{c,MILP},\dots,N_{p,MILP}-1\} \end{aligned} \tag{91}$$

With $\Delta_{u_2}(k+i) \geq 0$ the following constraint has to hold

$$u_{2,MILP}(k+i) - \underline{u}_2(k+i) \leq \Delta_{u_2}(k+i) \forall i \in \{0,\dots,N_{p,MILP}-1\} \tag{92}$$

Let $\underline{W}_{u_2}(k+i)$ be a weight. The penalty term penalizing positive deviations between $u_{2,MILP}$ and $\underline{u}_2$ added to (83) reads

$$J_{p,u_2}(k) = \sum_{i=0}^{N_{p,MILP}-1} \underline{W}_{u_2}(k+i) \cdot \Delta_{u_2}(k+i) \tag{93}$$

Note that if the MILP HVAC provides a smaller net mass flow of CO_2 than the plant MPC requests the resulting deviation is not penalized.

6.5. Penalizing deviations from the desired net mass flow of H_2O

The MILP model computes (3) by

$$\begin{aligned} u_{3,MILP}(k+i) = {} & \dot{m}_{ODA}(k+i) \\ & \cdot (\underline{x}_{H_2O,ODA}(k+i) - \underline{x}_{H_2O,IDA}(k+i)) \\ & - \dot{m}_{H_2O,COND}(k+i) \\ & \forall i \in \{0,\dots,N_{c,MILP}-1\} \end{aligned} \tag{94}$$

$$\begin{aligned} u_{3,MILP}(k+i) = {} & \dot{m}_{ODA}(k+i_f) \\ & \cdot (\underline{x}_{H_2O,ODA}(k+i) - \underline{x}_{H_2O,IDA}(k+i)) \\ & - \dot{m}_{H_2O,COND}(k+i) \\ & \text{with}\, i_f = N_{c,MILP}-1 \\ & \forall i \in \{N_{c,MILP},\dots,N_{p,MILP}-1\} \end{aligned} \tag{95}$$

The approach outlined in Section 6.4 is utilized to penalize positive deviations between $u_{3,MILP}$ and $\underline{u}_3$. However, also the approach outlined in Section 6.3 can be chosen to penalize positive and negative deviations between $u_{3,MILP}$ and $\underline{u}_3$.

6.6. Penalizing supply air temperature ϑ_{SUP} violations

The MILP considers the supply air temperature constraints by approximation as outlined in the following.

Assuming that water content of the supply air $x_{H_2O,SUP}$ equals the vapor content, the enthalpy of the supply air is computed by

$$h_{SUP} = (\underline{c}_{p,a} + x_{H_2O,SUP} \cdot \underline{c}_{p,v}) \cdot \vartheta_{SUP} + x_{H_2O,SUP} \cdot \underline{r}_0. \tag{96}$$

Thus the enthalpy of the supply air is a function of the supply air temperature ϑ_{SUP} and the water content of the supply air $x_{H_2O,SUP}$.

Neglecting here possible condensate, the water content of the supply air can be computed from (3) by

$$x_{H_2O,SUP} = \alpha \cdot \underline{x}_{H_2O,ODA} + (1-\alpha) \cdot \underline{x}_{H_2O,IDA} \tag{97}$$

The net enthalpy flow introduced to the plant due to the HVAC (i.e. the control variable u_1 of the linear MPC for the plant model, see Eq. (4)) is given by

$$\underline{u}_1 = \dot{m}_{SUP} \cdot (h_{SUP}(\vartheta_{SUP}, x_{H_2O,SUP}) - h_{IDA}) \tag{98}$$

Utilizing (98), (97) and (96) the supply air temperature ϑ_{SUP} as a function of the MILP decision variables is given by

$$\vartheta_{SUP} = \frac{\frac{\underline{u}_1}{\dot{m}_{SUP}} + \underline{h}_{IDA} - (\alpha\underline{x}_{H_2O,ODA} + (1-\alpha)\underline{x}_{H_2O,IDA})\underline{r}_0}{(\underline{c}_{p,a} + (\alpha\underline{x}_{H_2O,ODA} + (1-\alpha)\underline{x}_{H_2O,IDA})\underline{c}_{p,v})} \tag{99}$$

For a given desired net enthalpy flow $\underline{u}_1$, the given enthalpy of the indoor air $\underline{h}_{IDA}$, water content of the indoor $\underline{x}_{H_2O,IDA}$ and outdoor air $\underline{x}_{H_2O,ODA}$, and assumed constant parameters $\underline{r}_0, \underline{c}_{p,a}$, and $\underline{c}_{p,v}$, respectively, it is a nonlinear function in the decision variables $\dot{m}_{SUP}$ and α.

To avoid adding nonlinear constraints to the MILP model (and consequently the MILP model becoming a computationally expensive MINLP model) the constraints on the supply air temperature are modeled indirectly. Furthermore, to prevent that the optimization problem gets infeasible these constraints where modeled as soft constraints as outlined in the following. Note that $\delta_{\alpha,j} = 1$ implies that the fresh air ratio $\alpha = \underline{\alpha}_j$. For time instance $k+i$ the constraints due to $\underline{\vartheta}_{SUP,min}$ are

$$\begin{aligned} \underline{m}_{\Delta_{u_1,\vartheta_{SUP_{min}}v}}(1-\delta_{\alpha,j}) \leq {} & \Delta_{u_1,\vartheta_{SUP_{min}}v} \\ & - \left(\dot{m}_{SUP}(\underline{h}_{SUP,\vartheta_{SUP,min},j} - \underline{h}_{IDA}) - u_{1,MILP}\right) \end{aligned} \tag{100}$$

with

$$\underline{h}_{SUP,\vartheta_{SUP,min},j} = (\underline{c}_{p,a} + \underline{x}_{H_2O,SUP,j} \cdot \underline{c}_{p,v}) \cdot \underline{\vartheta}_{SUP,min} + \underline{x}_{H_2O,SUP,j} \cdot \underline{r}_0 \tag{101}$$

and the constraints due to $\underline{\vartheta}_{SUP,max}$ are

$$\begin{aligned} \underline{m}_{\Delta_{u_1,\vartheta_{SUP_{max}}v}} \cdot (1-\delta_{\alpha,j}) \leq {} & \Delta_{u_1,\vartheta_{SUP_{max}}v} \\ & - \left(u_{1,MILP} - \dot{m}_{SUP}(\underline{h}_{SUP,\vartheta_{SUP,max},j} - \underline{h}_{IDA})\right) \end{aligned} \tag{102}$$

with

$$\underline{h}_{\mathrm{SUP},\vartheta_{\mathrm{SUP,max}},j} = \left(\underline{c}_{\mathrm{p,a}} + \underline{x}_{\mathrm{H_2O,SUP},j} \cdot \underline{c}_{\mathrm{p,v}}\right) \cdot \underline{\vartheta}_{\mathrm{SUP,max}} + \underline{x}_{\mathrm{H_2O,SUP},j} \cdot \underline{r}_0. \tag{103}$$

In (100) and (102) the quantities $\underline{m}_{\Delta_{u_1,\vartheta_{\mathrm{SUP_{min}}}v}} \ll 0$ and $\underline{m}_{\Delta_{u_1,\vartheta_{\mathrm{SUP_{max}}}v}} \ll 0$ are appropriately chosen constants. Whereas the quantities $\Delta_{u_1,\vartheta_{\mathrm{SUP_{min}}}v}(k+i) \geq 0$ and $\Delta_{u_1,\vartheta_{\mathrm{SUP_{max}}}v}(k+i) \geq 0$ (for $i = 1,\ldots,N_{\mathrm{p,MILP}} - 1$) are auxiliary decision variables. In (101) and (103) $\underline{x}_{\mathrm{H_2O,SUP},j}$ is given by

$$\underline{x}_{\mathrm{H_2O,SUP},j} = \underline{\alpha}_j \cdot \underline{x}_{\mathrm{H_2O,ODA}} + (1 - \underline{\alpha}_j) \cdot \underline{x}_{\mathrm{H_2O,IDA}}. \tag{104}$$

With the weight $\underline{W}_{u_1,\vartheta_{\mathrm{SUP_{min}}}v}(k+i)$ the terms penalizing supply air temperature ϑ_{SUP} constraint violations by approximation added to (83) read:

$$J_{\mathrm{p},u_1,\vartheta_{\mathrm{SUP_{min}}}v}(k) = \sum_{i=0}^{N_{\mathrm{p,MILP}}-1} \underline{W}_{u_1,\vartheta_{\mathrm{SUP_{min}}}v}(k+i)\Delta_{u_1,\vartheta_{\mathrm{SUP_{min}}}v}(k+i) \tag{105}$$

$$J_{\mathrm{p},u_1,\vartheta_{\mathrm{SUP_{max}}}v}(k) = \sum_{i=0}^{N_{\mathrm{p,MILP}}-1} \underline{W}_{u_1,\vartheta_{\mathrm{SUP_{max}}}v}(k+i)\Delta_{u_1,\vartheta_{\mathrm{SUP_{max}}}v}(k+i) \tag{106}$$

6.7. Penalizing compressor on/off switching

I order to reduce switching, each time a refrigeration machine's compressor is switched from OFF to ON (or vice versa) a penalized term is added to the overall cost function (83). This is outlined in the following.

If at some time instance $k+i$ a compressor is switched from off to on the indicator

$$I_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i) = \delta_{\mathrm{AC,ON},c}(k+i) - \delta_{\mathrm{AC,ON},c}(k+i-1) \tag{107}$$

takes a value of 1, otherwise 0 or −1. With (107) and the auxiliary binary decision variable $\delta_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i)$ the following constraints have to hold (at time instance $k+i, \forall i \in \{0,\ldots,N_{\mathrm{c,MILP}}-1\}$):

$$I_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i) \leq \delta_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i) \tag{108}$$

$$-2 \cdot \left(1 - \delta_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i)\right) \leq I_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i) - 1 \tag{109}$$

$$I_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i) - 1 \leq -1 \cdot \left(1 - \delta_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i)\right) \tag{110}$$

With the user defined penalty weight $W_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i)$ the penalty term with respect to switching a compressor from OFF to ON reads

$$J_{\mathrm{p,AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k) = \sum_{i=0}^{N_{\mathrm{c,MILP}}-1} W_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i) \cdot \delta_{\mathrm{AC},\uparrow^{\mathrm{ON}}_{\mathrm{OFF}},c}(k+i). \tag{111}$$

6.8. Penalizing changing electrical compressor power/rotational speed

Depending on the utilized MILP model for the refrigeration machine (Section 5.5.1 or 5.5.2, respectively) either a change of the electrical power consumption ("ACS" and "ACD" model) or a change of the rotational speed ("ACV" model) of the compressor can be penalized. This is outlined for the "ACS" model (see Section 5.5.1):

Let $\underline{W}_{\mathrm{ACS,change},c}(k+i)$ be a weight. The penalty term penalizing changes of the electrical power consumption between consecutive time steps added to (83) reads

$$J_{\mathrm{p,ACS,change},c}(k) = \sum_{i=0}^{N_{\mathrm{c,MILP}}-1} \underline{W}_{\mathrm{ACS,change},c}(k+i) \cdot \left(P_{\mathrm{ACS},c}(k+i) - P_{\mathrm{ACS},c}(k+i-1)\right). \tag{112}$$

7. Simulation results

7.1. Case study 1: Viennese Ultra Low Floor tram: presented hierarchical framework vs. state of the art state machine

In this case study the power consumption and the performance of the HVAC of the Viennese Ultra Low Floor (ULF) tram is investigated by simulation. The control concept of the currently implemented controller, a state of the art finite state machine (FSM), is compared to the presented hierarchical MPC + MILP-MPC framework. The thermal plant model and the HVAC model of the Viennese ULF tram operated by its FSM was validated based on a detailed measurement campaign, see [17]. Background on FSM can be found in [25–27].

7.1.1. HVAC components of the Viennese ULF tram

The HVAC of the Viennese ULF tram is rather simple and has only a few number of degrees of freedom (DOF) for control. However, even with this low number of DOF its rule based controller has 193 transitions (rules). These rules are the result of expert knowledge and time-consuming trial-and-error tests in a climatic chamber.

The mixing chamber of the HVAC can provide two different fresh air ratios α. The compressor of its refrigeration machine can be turned on or off. The cooling power can be reduced by opening a by-pass valve. The heater battery and the supply air fan both have three operating stages.

7.1.2. Simulation settings and results

A typical acceptance test conducted in a climatic chamber is simulated in the following. The outdoor temperature is increased and decreased between 0 °C and 30 °C (temperature gradient 3 °C/h). One vehicle side is exposed to radiation (600 W/m^2; incidence angle 30° to the horizontal). Occupancy is 42 persons. Since the FSM is a non-predictive controller, for a fair comparison of the two control concepts $N_{\mathrm{p}} = N_{\mathrm{c}} = N_{\mathrm{p,MILP}} = N_{\mathrm{c,MILP}} = 1$ is chosen for the MPC + MILP-MPC framework. The respective results are denoted by "MPC + MILP" in this case. Note that this choice effectively eliminates the predictive nature but retains the other features of the presented framework. Fig. 4 compares the results in terms of the indoor air temperature. As evident from Fig. 4 with the rule based FSM larger deviations from the set point temperature occur. The HVAC control strategies are compared in Fig. 5. Both control concepts observe the minimum consecutive on/off constraints of the compressor. However, control action of the MPC + MILP is higher. Depending on the requirements compressor switching could be reduced by increasing the respective weights (see Section 6.7). The MPC + MILP utilizes "free cooling" by optimally varying the fresh air ratio α. With the FSM the overall energy

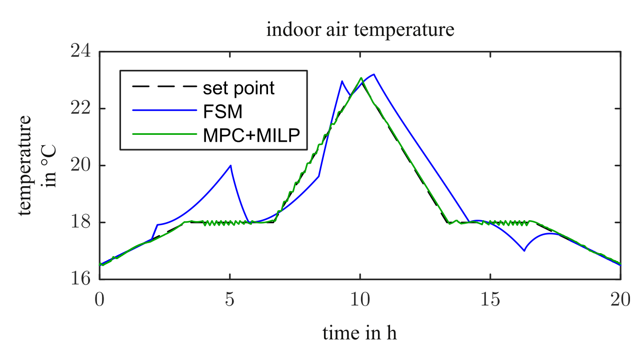


Fig. 4. Case study 1: indoor air temperature.

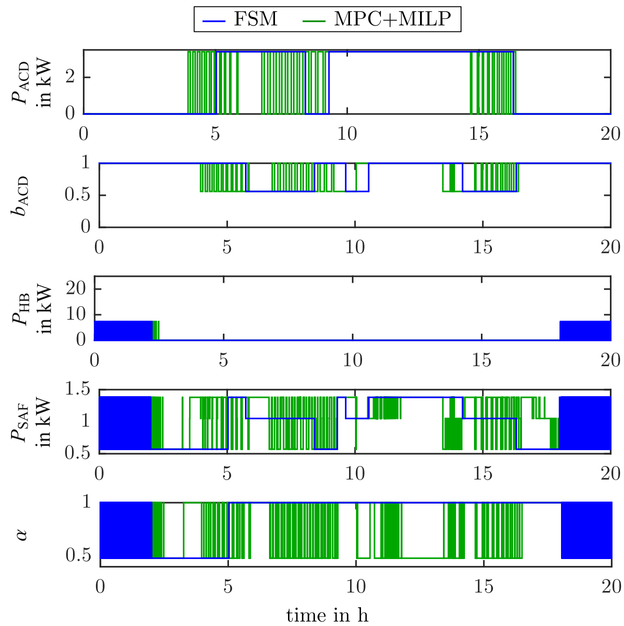


Fig. 5. Case study 1: HVAC control strategy.

consumption of the HVAC during the test is 66.44 kWh, whereas with the MPC + MILP it is only 61.66 kWh.

Fig. 6 shows a simulation where the dead band for the MPC + MILP is the area between the indoor air temperature achieved by the "FSM" and the set point. In this case the MPC + MILP's energy consumption is only 60.34 kWh. This is a further reduction in energy consumption by 2.14%.

Setting $N_p \geq N_c \geq N_{p,MILP} \geq N_{c,MILP} > 1$ for the hierarchical MPC + MILP-MPC framework, performance, switching frequency, and energy consumption can be even more improved.

7.2. Case study 2: simultaneous control of temperature, CO_2 level, and humidity

This second case study demonstrates the ability of the MIMO MPC + MILP-MPC framework to control temperature, CO_2 level, and humidity simultaneously.

7.2.1. HVAC configuration

The HVAC of this second case study has a refrigeration machine (RM) with a variable speed compressor (see Section 5.5.1). Since frequency variable operation is only possible within a certain range its electrical power consumption P_{ACS} is a semi-continuous decision variable. The cooling power of this RM can be additionally adjusted (reduced) by opening a by-pass valve. The minimum consecutive on/off time of the RM is $T_{ACS,ON} = 100$ s/$T_{ACS,OFF} = 40$ s. The supply air fan has 6 switching stages. The mixing chamber can provide 8 different fresh air ratios α and the heating power of the heater battery can be varied continuously.

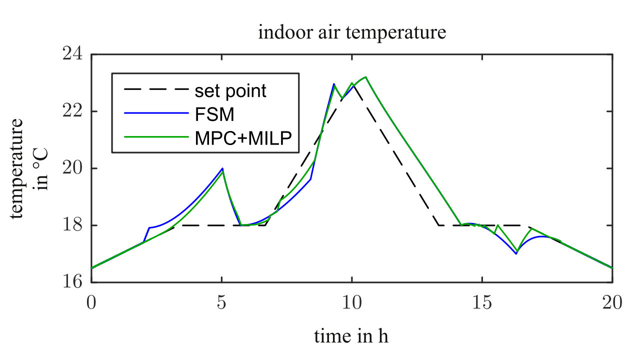


Fig. 6. Case study 1: indoor air temperature when dead band is the area between the indoor air temperature achieved by the "FSM" and the set point.

7.2.2. Simulation settings and results

In the following a couple of step changes are performed. It is not claimed that these rapid changes are realistic. The aim is to show an illustrative simulation where each time step is clearly visible. The presented controller can handle even these extreme requirements. The outdoor air is cold (15 °C), has a high water content (10.24 g/kg), and a low CO_2 level (400 ppmv). Occupancy is 42 persons. CO_2 production rate per person is 0.7 g_{CO_2}/ min. The mass of water due to one person's perspiration depends on the indoor air temperature and varies around 0.48 g_{H2O}/ min.

Fig.7 shows the indoor air temperature ϑ_{IDA} and step changes of its prescribed upper/lower limit. Fig. 8 shows the CO_2 level of the indoor air c_{IDA} and its upper limit step changes (no desired value is given). Fig.9 shows the water content of the indoor air $x_{H_2O,IDA}$ and step changes of its prescribed upper/lower limit. Fig. 10 shows the HVAC control strategy.

From Figs. 7–9 it can be seen that the proposed concept effectively decouples the three control variables and achieves high tracking performance only limited by the actuators.

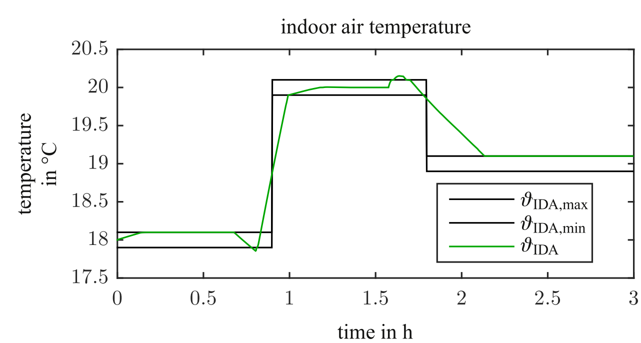


Fig. 7. Case study 2: indoor air temperature.

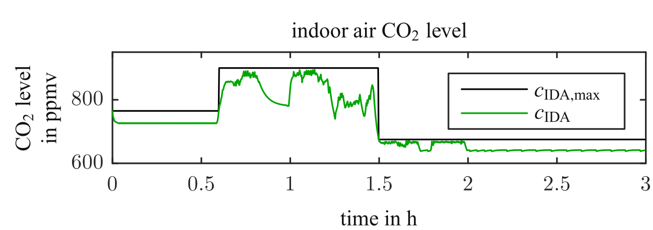


Fig. 8. Case study 2: indoor air CO_2 level.

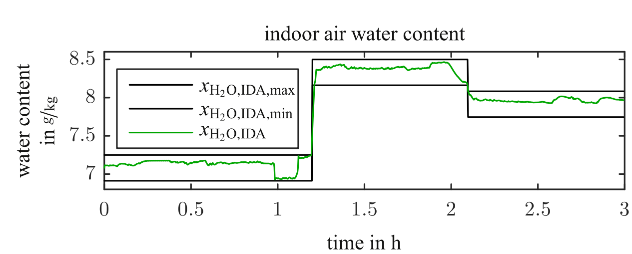


Fig. 9. Case study 2: indoor air water content.

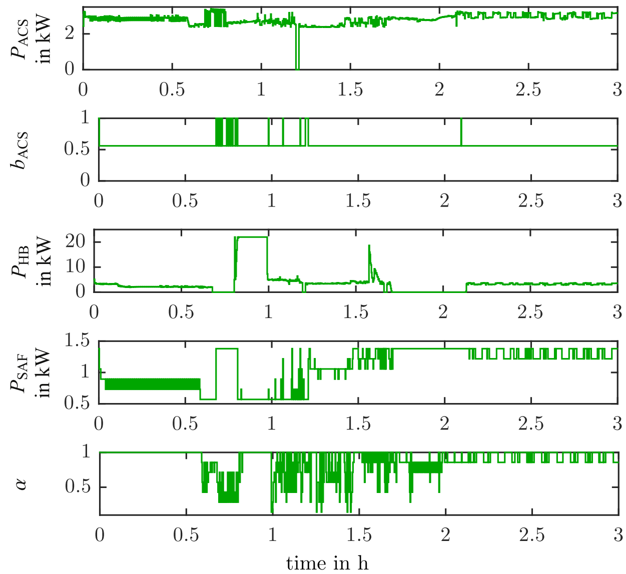


Fig. 10. Case study 2: HVAC control strategy.

7.3. Case study 3: non predictive MILP vs. predictive MILP-MPC

In this third case study it is shown that performance, switching moment, and energy consumption are improved by considering more than only the first sample of the future control trajectory $\boldsymbol{U}(k) = [\boldsymbol{u}^{\mathrm{T}}(k), \ldots, \boldsymbol{u}^{\mathrm{T}}(k + N_c - 1)]^{\mathrm{T}}$ of the plant MPC in the decision upon the optimal HVAC operation at the present moment k. In other words the third case study compares the case $N_p \geq N_c \geq N_{p,\mathrm{MILP}} = N_{c,\mathrm{MILP}} = 1$ with the case $N_p \geq N_c \geq N_{p,\mathrm{MILP}} \geq N_{c,\mathrm{MILP}} > 1$.

A situation is simulated where on a hot and sunny summer day the plant is first exposed to high ambient temperature and radiation (35 °C; 600 W/m²) followed by a step change to low ambient temperature and no radiation (10 °C; 0 W/m²). Such a situation could occur to a vehicle (e.g. a metro) entering a tunnel passage, or because of a sudden summer thunderstorm. It is assumed that predictions of ambient temperature and radiation are available from a look-ahead system utilizing position information, or the weather forecast, respectively. Instead of only one refrigeration machine (RM) the HVAC of this third case study has three RMs, see Fig. 11.

Two are equal and small (ACV, 1 and ACV, 2) with small compressors. These compressors have small minimum consecutive on/off times ($T_{\mathrm{ACV,1,ON}} = T_{\mathrm{ACV,2,ON}} = 100$ s, $T_{\mathrm{ACV,1,OFF}} = T_{\mathrm{ACV,2,OFF}} = 40$ s). Additionally, there is one large RM (ACV, 3) with a powerful compressor for which large minimum consecutive on/off times have to be observed ($T_{\mathrm{ACV,3,ON}} = 800$ s, $T_{\mathrm{ACV,3,OFF}} = 240$ s). The compressors of the small RMs can be operated with four rotational speeds. Whereas the compressor of the large RM can only be turned on and off. The small ones and the large ones have each a semi-continuous operable electronic expansion valve to adjust their cooling power. The motor of the supply air fan is frequency variable. The mixing chamber can provide two different fresh air ratios α and the heater battery has three switching stages. In all the HVAC has 1536 discrete states and 4 continuous states.

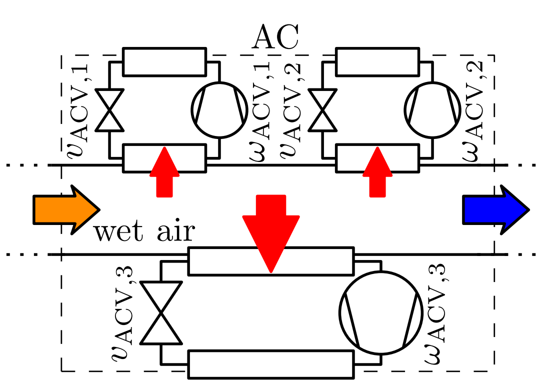


Fig. 11. Case study 3: conceptual refrigeration machines.

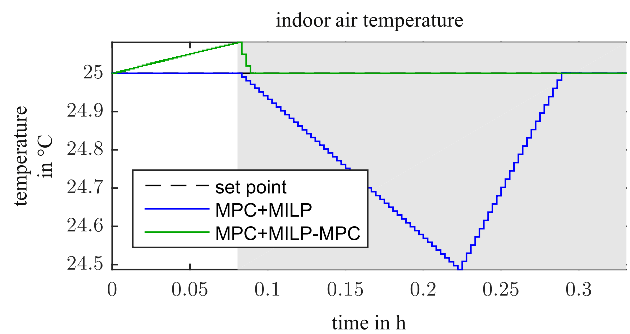


Fig. 12. Case study 3: indoor air temperature.

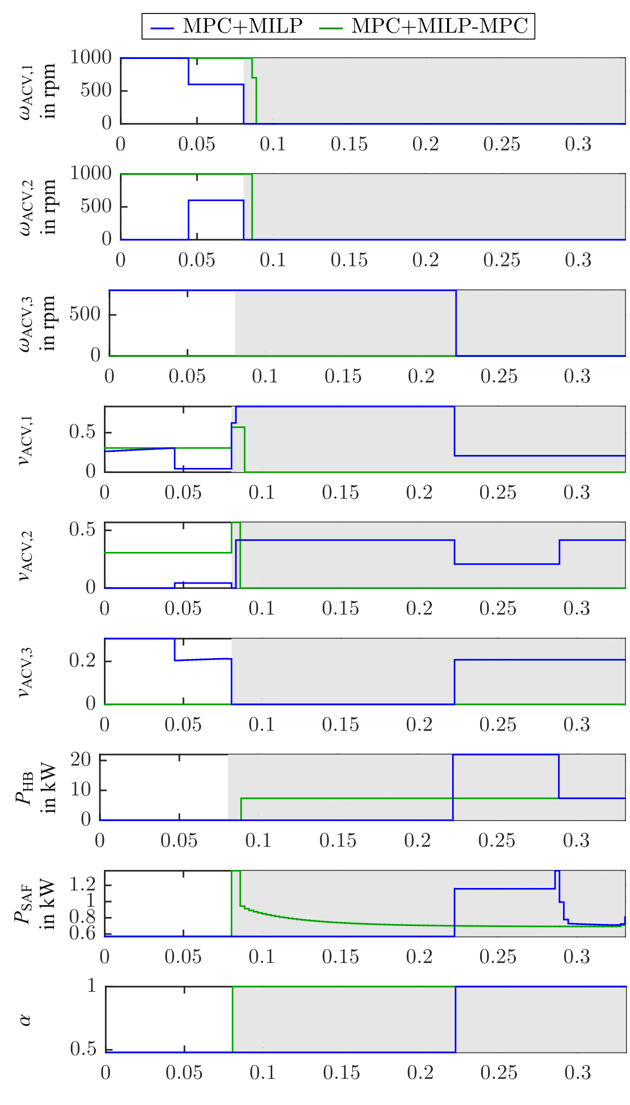


Fig. 13. Case study 3: HVAC control strategy.

Figs. 12 and Fig. 13 show the results of the third case study in terms of indoor air temperature and HVAC control strategy, respectively. The results denoted by "MPC + MILP" were obtained with $N_\mathrm{p} = N_\mathrm{c} = 82$ and $N_{\mathrm{p,MILP}} = N_{\mathrm{c,MILP}} = 1$. Whereas the results denoted by "MPC + MILP-MPC" were obtained with $N_\mathrm{p} = N_\mathrm{c} = N_{\mathrm{p,MILP}} = N_{\mathrm{c,MILP}} = 82$ (i.e. also the "MILP-MPC" oversees the minimum consecutive on/off times of the compressors). All other settings were the same. A sampling time of $T_\mathrm{s} = 10$ s was used. In Figs. 12 and 13 the period with low ambient temperature and no radiation is shaded in gray. At the beginning in both cases the indoor air temperature equals its set point (see Fig. 12). As evident from Fig. 13 the non predictive MILP controller (case: "MPC + MILP") optimizing the HVAC operation turns on all three compressors. Whereas the predictive MILP-MPC (case: "MPC + MILP-MPC") does not turn on the large compressor with the large minimum consecutive on time $T_{\mathrm{ACV,3,ON}} = 800$ s. After the step change the thermal load soon becomes negative, i.e. requiring heating. Since the large compressor once turned on can not be turned off for a long time, the temperature in the case "MPC + MILP" keeps falling under the set point temperature until the large compressor finally can be switched off. As evident from Fig. 12 the overall temperature performance in the case "MPC + MILP-MPC" is better. Furthermore the case "MPC + MILP-MPC" requires less compressor switching and because less cooling and counter-heating is applied it requires less energy consumption for the HVAC. Simultaneously heating and cooling was not allowed in both cases. As evident from this case study better results are obtained if also the HVAC operation optimization (i.e. the MILP) oversees the latency periods of the HVAC components.

8. Conclusion

A unified method for energy-, wear-, and comfort-optimized operation of HVACs, based on a flexible and modular MILP model of HVACs, is presented. A hierarchical MIMO MPC framework is proposed. On the upper-level a computationally inexpensive plant MPC with large control and prediction horizon takes care of the slow dynamics of the (thermal) plant. It takes all available disturbance predictions into account. On the lower-level a MILP-MPC based on a MILP model of an HVAC system optimizes the HVAC operation. As shown by a case study its prediction horizon should at least oversee the latency periods of the HVAC components with dwell times.

On the upper-level also a classical PID controller could be applied. However, the increasing availability of accurate disturbance forecasts (like ambient temperature, occupancy, driving speed, etc.) motivate the usage of an MPC in the upper-level which can attain extra performance by utilizing this information. Moreover, by this means, also the lower-level controller can be an MPC. As demonstrated in this work the lower-level MPC can utilize the first $N_{\mathrm{p,MILP}}$ samples ($N_{\mathrm{p,MILP}} \leq N_\mathrm{c}$) of the future control trajectory $\boldsymbol{u}(k+i)$ (for $i = 0, \ldots, N_\mathrm{c} - 1$)) of the upper-level MPC to consider future demand of net enthalpy and mass flows. In this way the optimal switching moment of HVAC components, for which minimum consecutive on/off constraints have to be fulfilled, can be achieved.

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